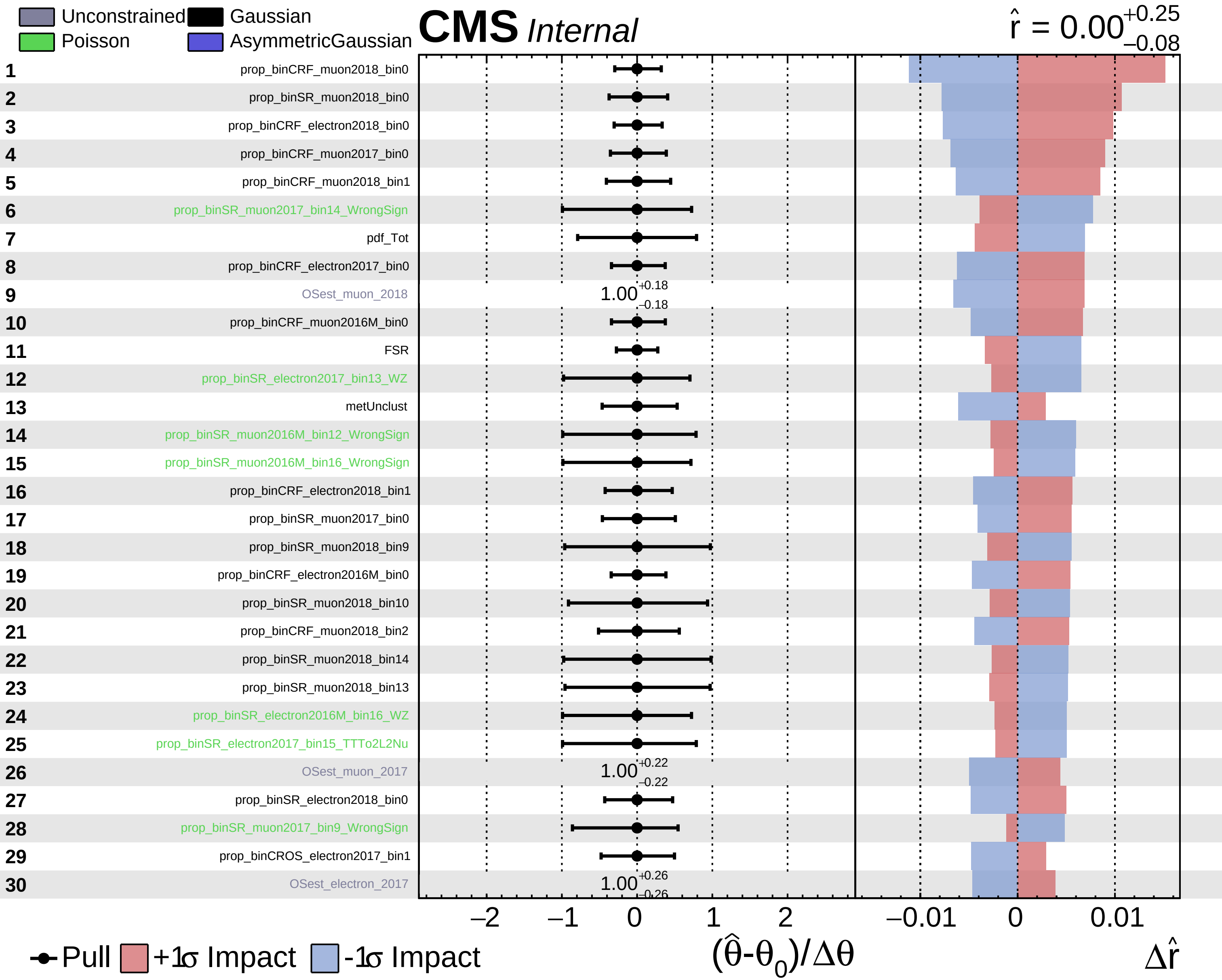
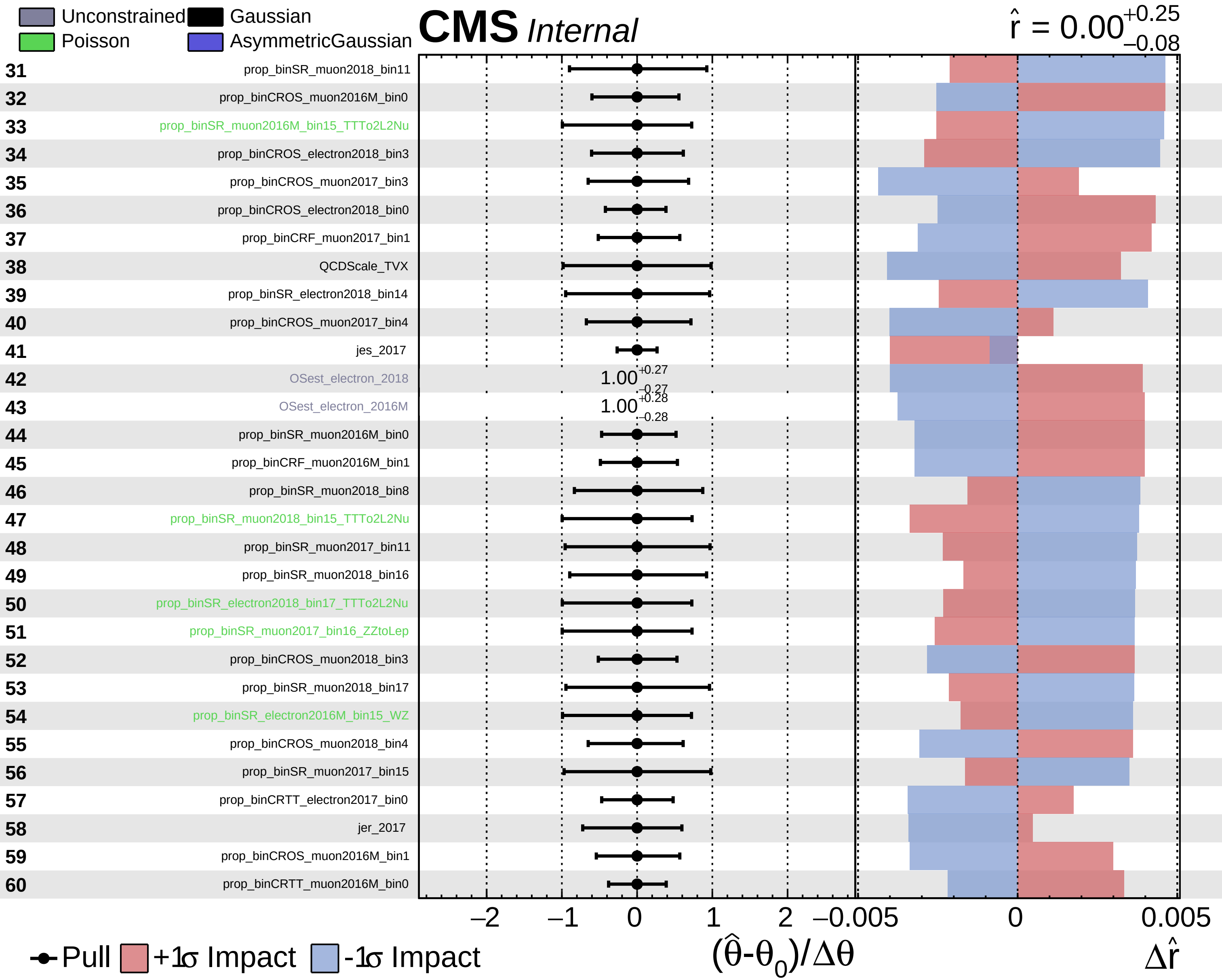
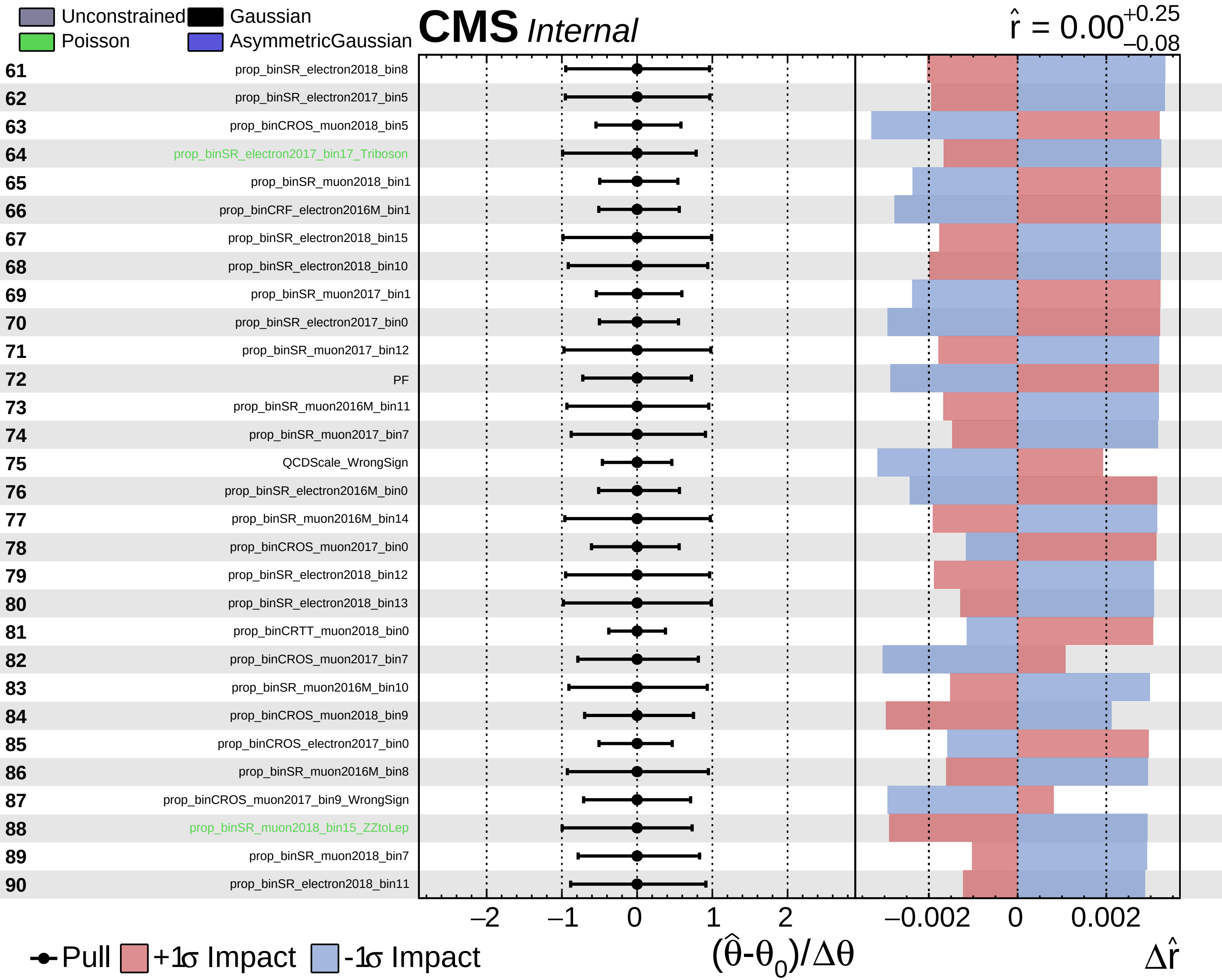
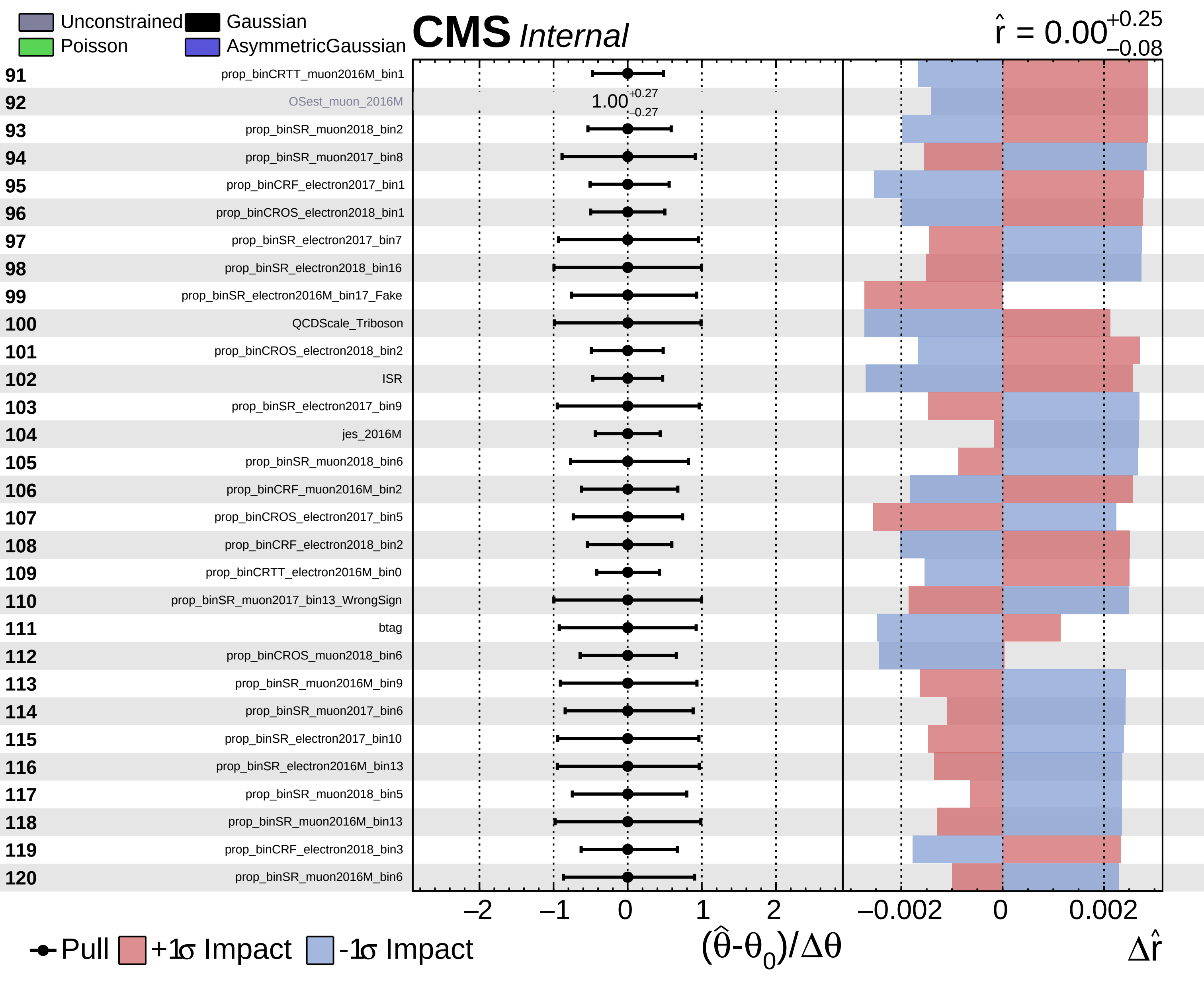

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
1 prop_binCRF_muon2018_bin0
2 prop_binSR_muon2018_bin0
3 prop_binCRF_electron2018_bin0
4 prop_binCRF_muon2017_bin0
5 prop_binCRF_muon2018_bin1
6 prop_binSR_muon2017_bin14_WrongSign
7 pdf_Tot
8 prop_binCRF_electron2017_bin0
9 OSest_muon_2018
$1.00^{+0.18}_{-0.18}$
10 prop_binCRF_muon2016M_bin0
11 FSR
12 prop_binSR_electron2017_bin13_WZ
13 metUnclust
14 prop_binSR_muon2016M_bin12_WrongSign
15 prop_binSR_muon2016M_bin16_WrongSign
16 prop_binCRF_electron2018_bin1
17 prop_binSR_muon2017_bin0
18 prop_binSR_muon2018_bin9
19 prop_binCRF_electron2016M_bin0
20 prop_binSR_muon2018_bin10
21 prop_binCRF_muon2018_bin2
22 prop_binSR_muon2018_bin14
23 prop_binSR_muon2018_bin13
24 prop_binSR_electron2016M_bin16_WZ
25 prop_binSR_electron2017_bin15_TTTo2L2Nu
26 OSest_muon_2017
$1.00^{+0.22}_{-0.22}$
27 prop_binSR_electron2018_bin0
28 prop_binSR_muon2017_bin9_WrongSign
29 prop_binCROS_electron2017_bin1
30 OSest_electron_2017
$1.00^{+0.26}_{-0.26}$
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

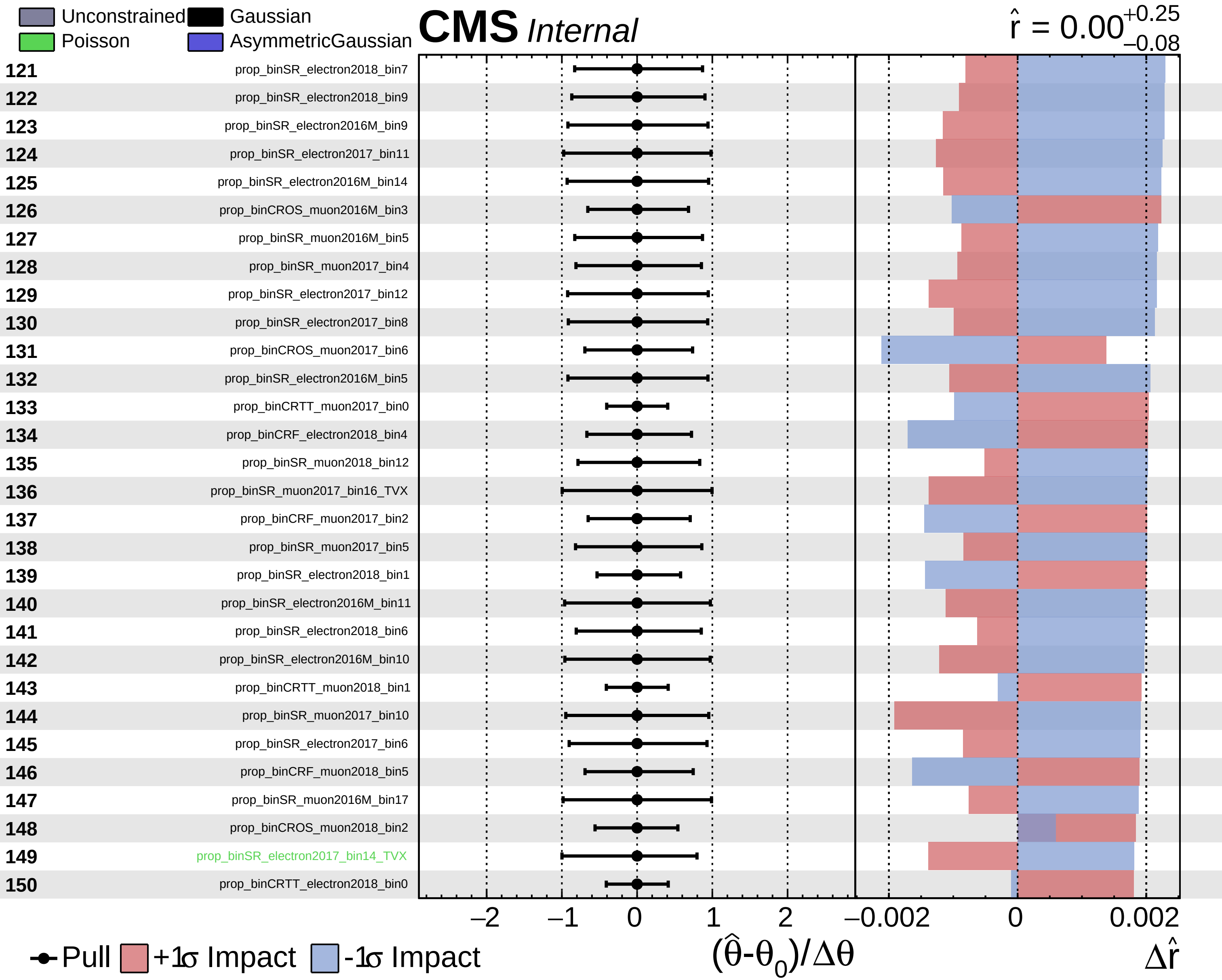


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.25 −0.08
31 prop_binSR_muon2018_bin11
32 prop_binCROS_muon2016M_bin0
33 prop_binSR_muon2016M_bin15_TTTo2L2Nu
34 prop_binCROS_electron2018_bin3
35 prop_binCROS_muon2017_bin3
36 prop_binCROS_electron2018_bin0
37 prop_binCRF_muon2017_bin1
38 QCDScale_TVX
39 prop_binSR_electron2018_bin14
40 prop_binCROS_muon2017_bin4
41 jes_2017
42 OSest_electron_2018
1.00 +0.27 −0.27
43 OSest_electron_2016M
1.00 +0.28 −0.28
44 prop_binSR_muon2016M_bin0
45 prop_binCRF_muon2016M_bin1
46 prop_binSR_muon2018_bin8
47 prop_binSR_muon2018_bin15_TTTo2L2Nu
48 prop_binSR_muon2017_bin11
49 prop_binSR_muon2018_bin16
50 prop_binSR_electron2018_bin17_TTTo2L2Nu
51 prop_binSR_muon2017_bin16_ZZtoLep
52 prop_binCROS_muon2018_bin3
53 prop_binSR_muon2018_bin17
54 prop_binSR_electron2016M_bin15_WZ
55 prop_binCROS_muon2018_bin4
56 prop_binSR_muon2017_bin15
57 prop_binCRTT_electron2017_bin0
58 jer_2017
59 prop_binCROS_muon2016M_bin1
60 prop_binCRTT_muon2016M_bin0
−2 −1 0 1 2 −0.005 0 0.005
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

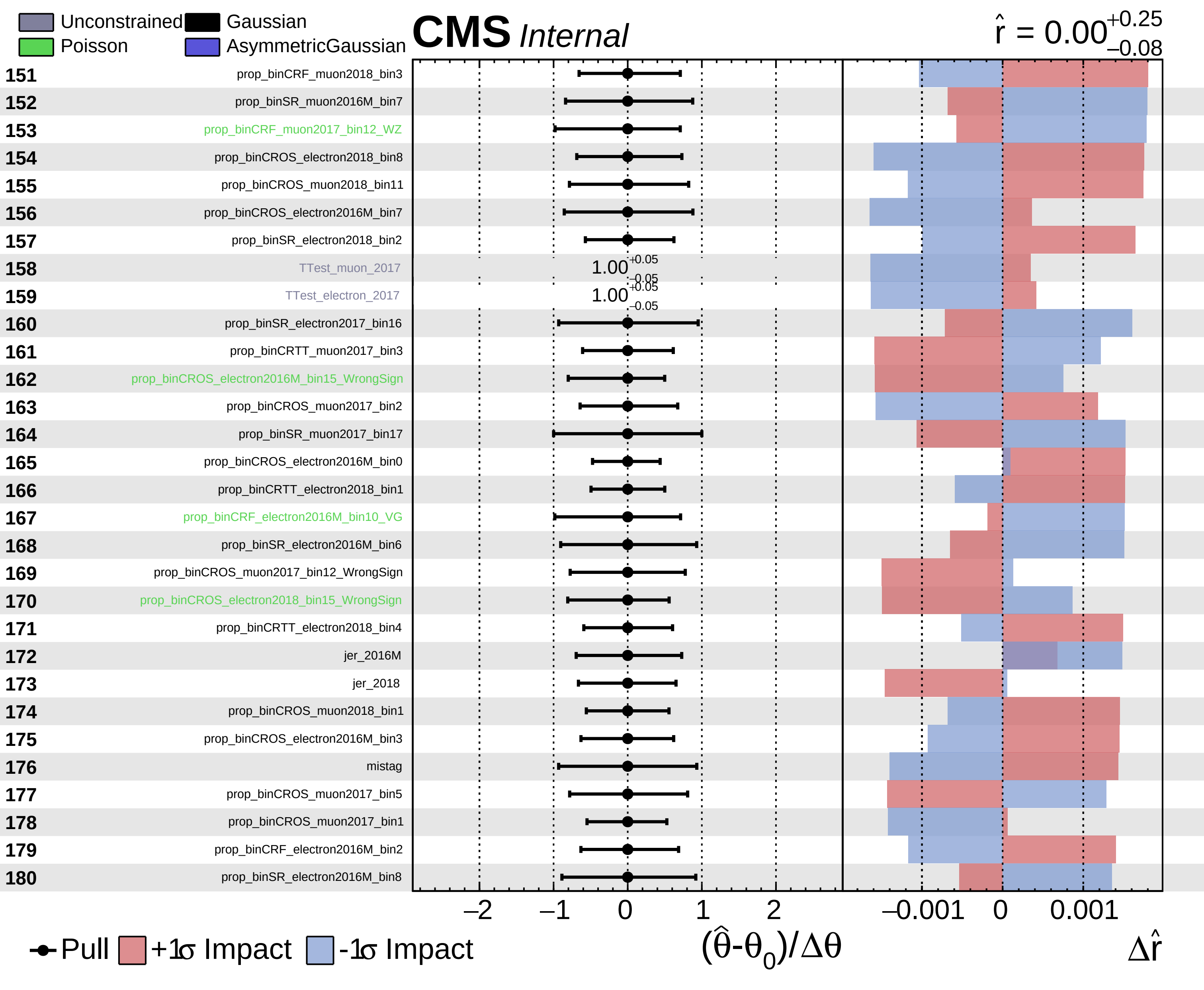

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
61 prop_binSR_electron2018_bin8
62 prop_binSR_electron2017_bin5
63 prop_binCROS_muon2018_bin5
64 prop_binSR_electron2017_bin17_Triboson
65 prop_binSR_muon2018_bin1
66 prop_binCRF_electron2016M_bin1
67 prop_binSR_electron2018_bin15
68 prop_binSR_electron2018_bin10
69 prop_binSR_muon2017_bin1
70 prop_binSR_electron2017_bin0
71 prop_binSR_muon2017_bin12
72 PF
73 prop_binSR_muon2016M_bin11
74 prop_binSR_muon2017_bin7
75 QCDScale_WrongSign
76 prop_binSR_electron2016M_bin0
77 prop_binSR_muon2016M_bin14
78 prop_binCROS_muon2017_bin0
79 prop_binSR_electron2018_bin12
80 prop_binSR_electron2018_bin13
81 prop_binCRTT_muon2018_bin0
82 prop_binCROS_muon2017_bin7
83 prop_binSR_muon2016M_bin10
84 prop_binCROS_muon2018_bin9
85 prop_binCROS_electron2017_bin0
86 prop_binSR_muon2016M_bin8
87 prop_binCROS_muon2017_bin9_WrongSign
88 prop_binSR_muon2018_bin15_ZZtoLep
89 prop_binSR_muon2018_bin7
90 prop_binSR_electron2018_bin11
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



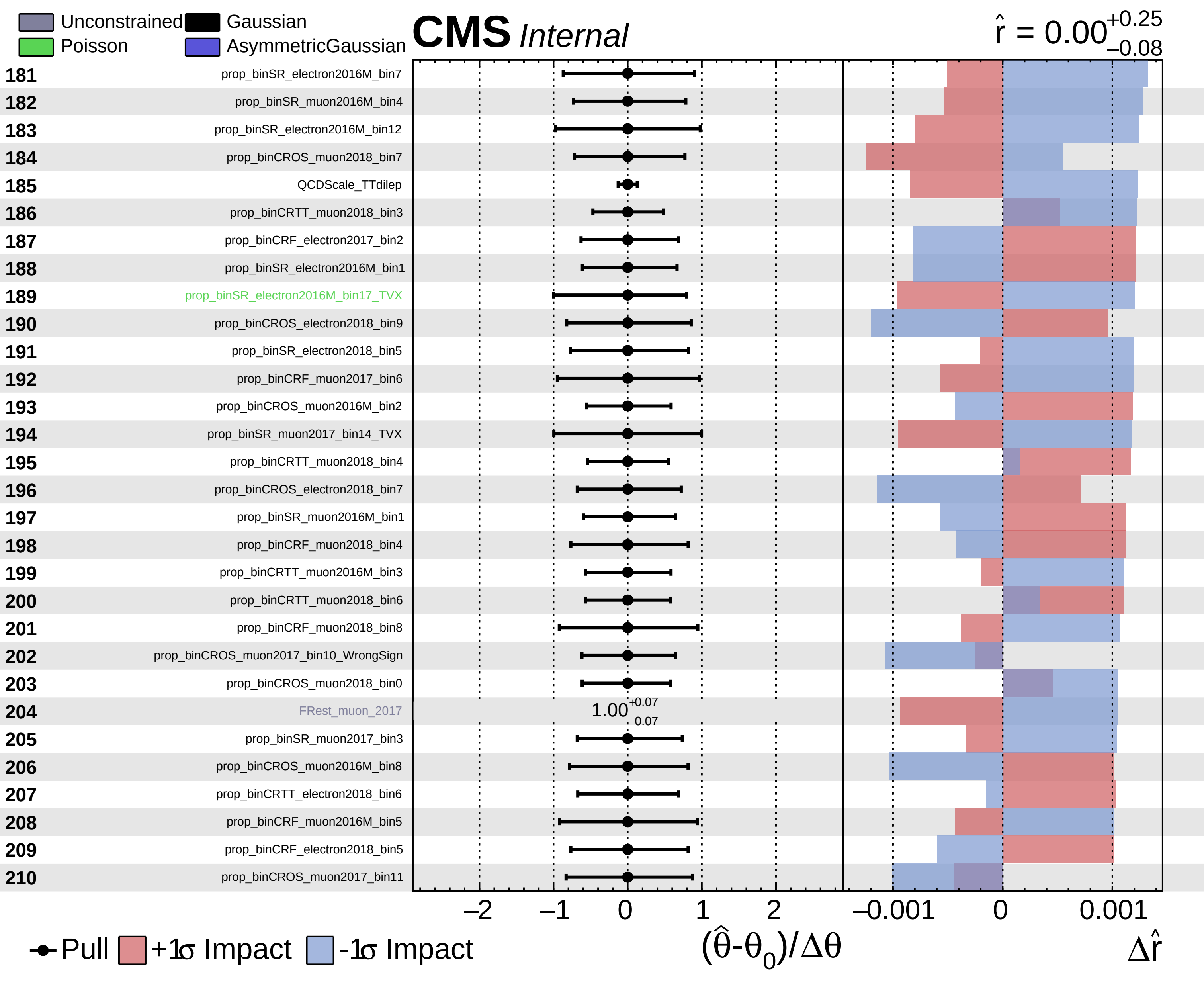
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
91 prop_binCRTT_muon2016M_bin1
92 OSest_muon_2016M
1.00 +0.27 −0.27
93 prop_binSR_muon2018_bin2
94 prop_binSR_muon2017_bin8
95 prop_binCRF_electron2017_bin1
96 prop_binCROS_electron2018_bin1
97 prop_binSR_electron2017_bin7
98 prop_binSR_electron2018_bin16
99 prop_binSR_electron2016M_bin17_Fake
100 QCDScale_Triboson
101 prop_binCROS_electron2018_bin2
102 ISR
103 prop_binSR_electron2017_bin9
104 jes_2016M
105 prop_binSR_muon2018_bin6
106 prop_binCRF_muon2016M_bin2
107 prop_binCROS_electron2017_bin5
108 prop_binCRF_electron2018_bin2
109 prop_binCRTT_electron2016M_bin0
110 prop_binSR_muon2017_bin13_WrongSign
111 btag
112 prop_binCROS_muon2018_bin6
113 prop_binSR_muon2016M_bin9
114 prop_binSR_muon2017_bin6
115 prop_binSR_electron2017_bin10
116 prop_binSR_electron2016M_bin13
117 prop_binSR_muon2018_bin5
118 prop_binSR_muon2016M_bin13
119 prop_binCRF_electron2018_bin3
120 prop_binSR_muon2016M_bin6
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

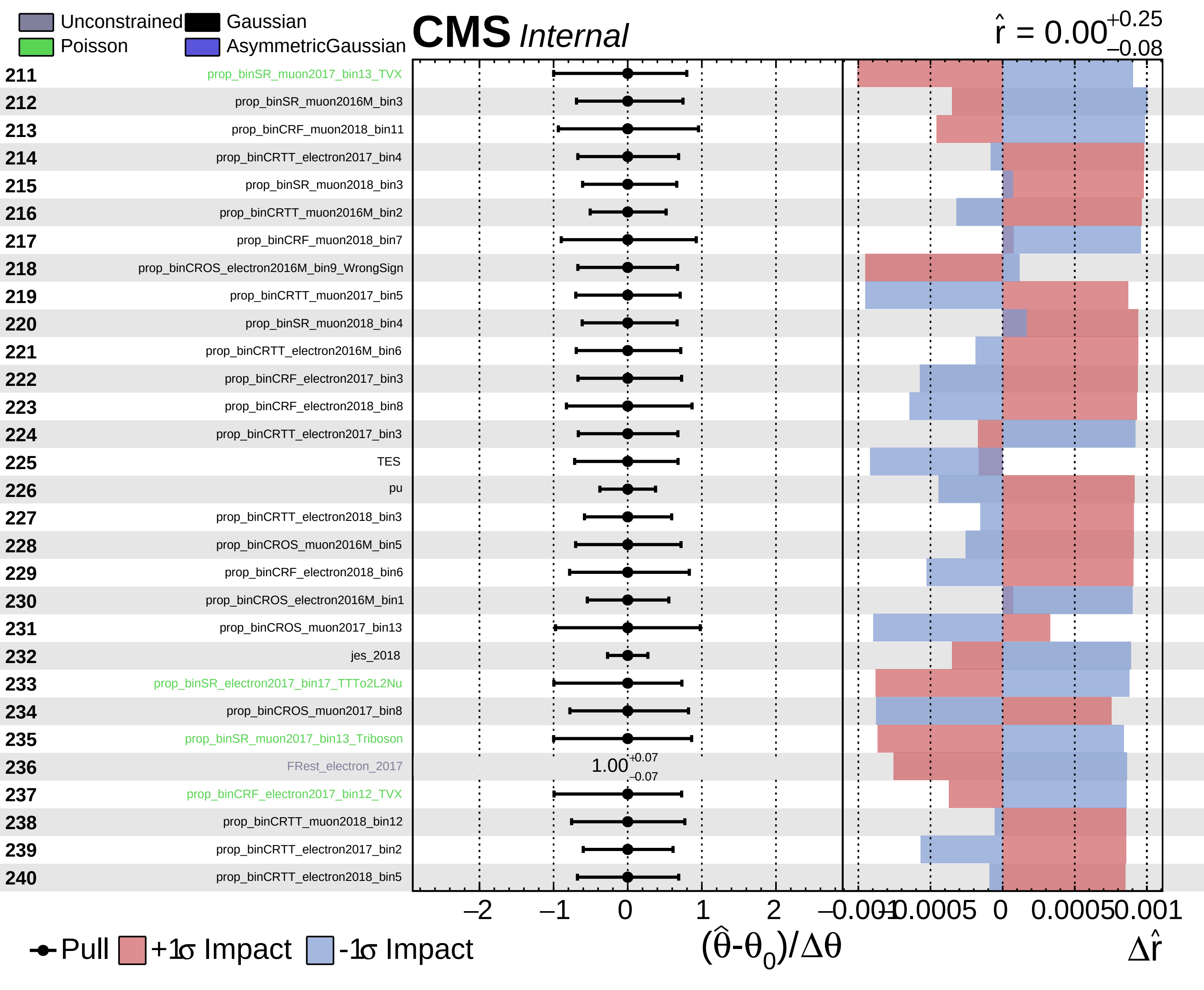


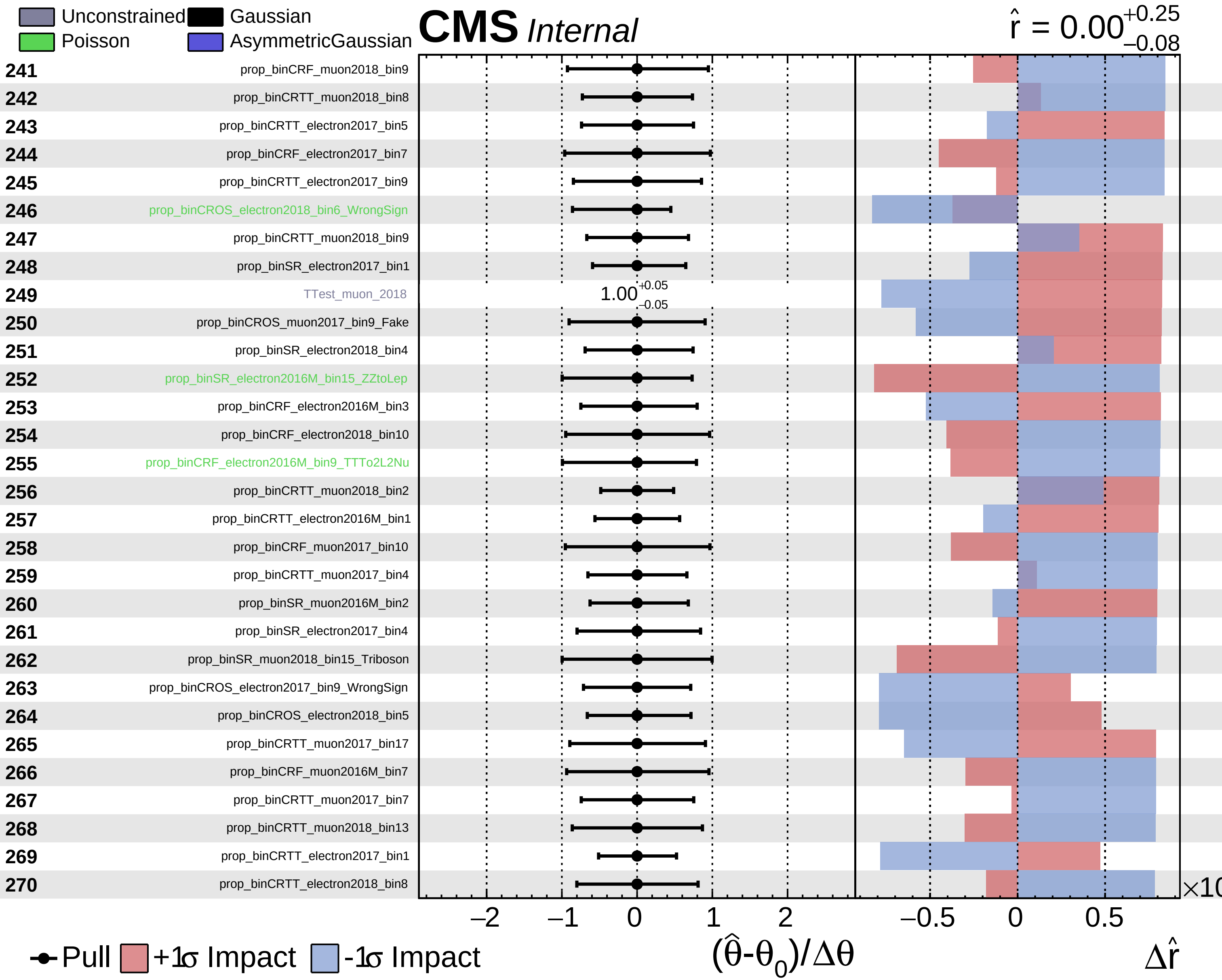
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
121 prop_binSR_electron2018_bin7
122 prop_binSR_electron2018_bin9
123 prop_binSR_electron2016M_bin9
124 prop_binSR_electron2017_bin11
125 prop_binSR_electron2016M_bin14
126 prop_binCROS_muon2016M_bin3
127 prop_binSR_muon2016M_bin5
128 prop_binSR_muon2017_bin4
129 prop_binSR_electron2017_bin12
130 prop_binSR_electron2017_bin8
131 prop_binCROS_muon2017_bin6
132 prop_binSR_electron2016M_bin5
133 prop_binCRTT_muon2017_bin0
134 prop_binCRF_electron2018_bin4
135 prop_binSR_muon2018_bin12
136 prop_binSR_muon2017_bin16_TVX
137 prop_binCRF_muon2017_bin2
138 prop_binSR_muon2017_bin5
139 prop_binSR_electron2018_bin1
140 prop_binSR_electron2016M_bin11
141 prop_binSR_electron2018_bin6
142 prop_binSR_electron2016M_bin10
143 prop_binCRTT_muon2018_bin1
144 prop_binSR_muon2017_bin10
145 prop_binSR_electron2017_bin6
146 prop_binCRF_muon2018_bin5
147 prop_binSR_muon2016M_bin17
148 prop_binCROS_muon2018_bin2
149 prop_binSR_electron2017_bin14_TVX
150 prop_binCRTT_electron2018_bin0
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



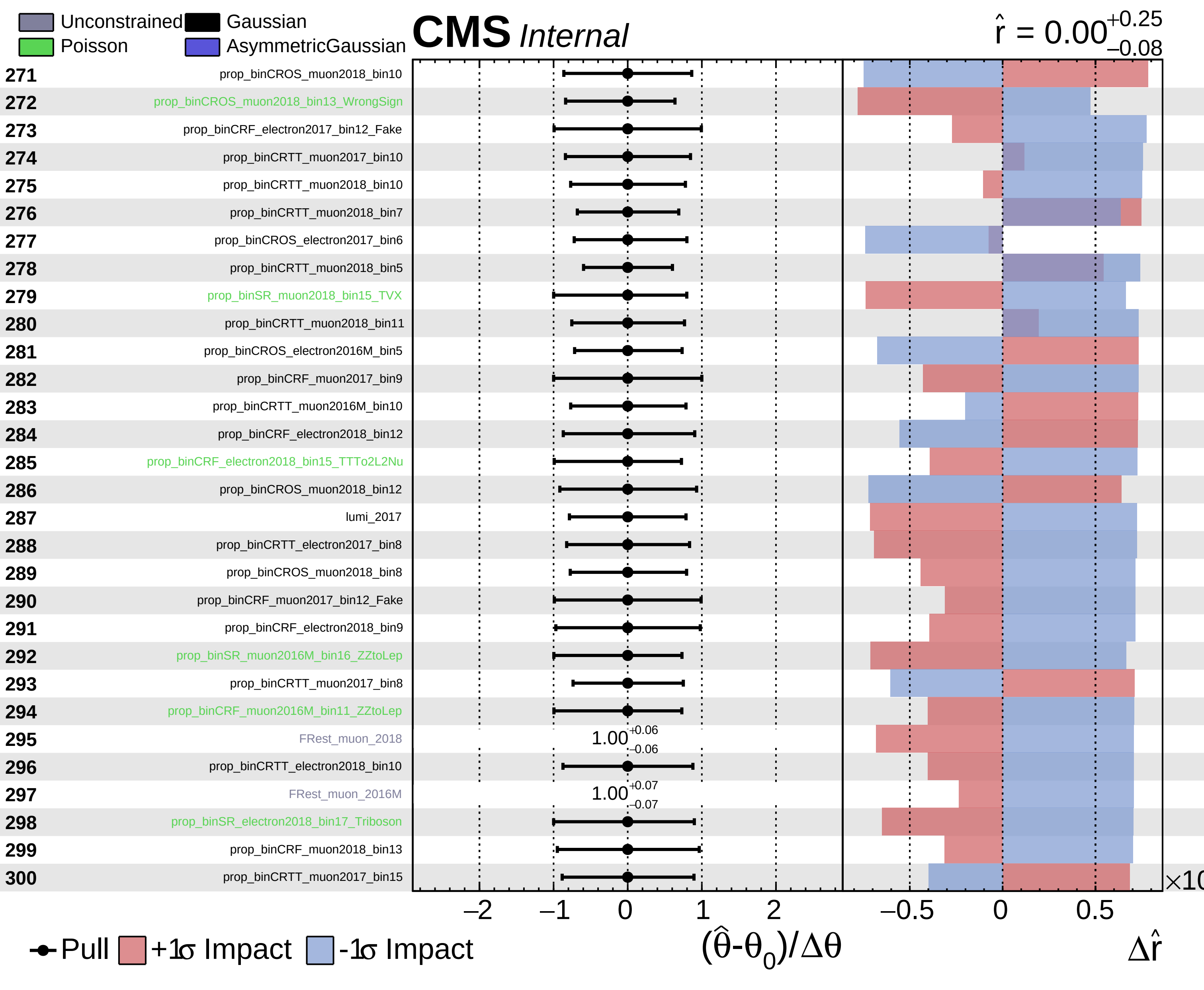
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
151 prop_binCRF_muon2018_bin3
152 prop_binSR_muon2016M_bin7
153 prop_binCRF_muon2017_bin12_WZ
154 prop_binCROS_electron2018_bin8
155 prop_binCROS_muon2018_bin11
156 prop_binCROS_electron2016M_bin7
157 prop_binSR_electron2018_bin2
158 TTest_muon_2017
$1.00^{+0.05}_{-0.05}$
159 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
160 prop_binSR_electron2017_bin16
161 prop_binCRTT_muon2017_bin3
162 prop_binCROS_electron2016M_bin15_WrongSign
163 prop_binCROS_muon2017_bin2
164 prop_binSR_muon2017_bin17
165 prop_binCROS_electron2016M_bin0
166 prop_binCRTT_electron2018_bin1
167 prop_binCRF_electron2016M_bin10_VG
168 prop_binSR_electron2016M_bin6
169 prop_binCROS_muon2017_bin12_WrongSign
170 prop_binCROS_electron2018_bin15_WrongSign
171 prop_binCRTT_electron2018_bin4
172 jer_2016M
173 jer_2018
174 prop_binCROS_muon2018_bin1
175 prop_binCROS_electron2016M_bin3
176 mistag
177 prop_binCROS_muon2017_bin5
178 prop_binCROS_muon2017_bin1
179 prop_binCRF_electron2016M_bin2
180 prop_binSR_electron2016M_bin8
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.25 −0.08
181 prop_binSR_electron2016M_bin7
182 prop_binSR_muon2016M_bin4
183 prop_binSR_electron2016M_bin12
184 prop_binCROS_muon2018_bin7
185 QCDScale_TTdilep
186 prop_binCRTT_muon2018_bin3
187 prop_binCRF_electron2017_bin2
188 prop_binSR_electron2016M_bin1
189 prop_binSR_electron2016M_bin17_TVX
190 prop_binCROS_electron2018_bin9
191 prop_binSR_electron2018_bin5
192 prop_binCRF_muon2017_bin6
193 prop_binCROS_muon2016M_bin2
194 prop_binSR_muon2017_bin14_TVX
195 prop_binCRTT_muon2018_bin4
196 prop_binCROS_electron2018_bin7
197 prop_binSR_muon2016M_bin1
198 prop_binCRF_muon2018_bin4
199 prop_binCRTT_muon2016M_bin3
200 prop_binCRTT_muon2018_bin6
201 prop_binCRF_muon2018_bin8
202 prop_binCROS_muon2017_bin10_WrongSign
203 prop_binCROS_muon2018_bin0
204 FRest_muon_2017
1.00 +0.07 −0.07
205 prop_binSR_muon2017_bin3
206 prop_binCROS_muon2016M_bin8
207 prop_binCRTT_electron2018_bin6
208 prop_binCRF_muon2016M_bin5
209 prop_binCRF_electron2018_bin5
210 prop_binCROS_muon2017_bin11
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

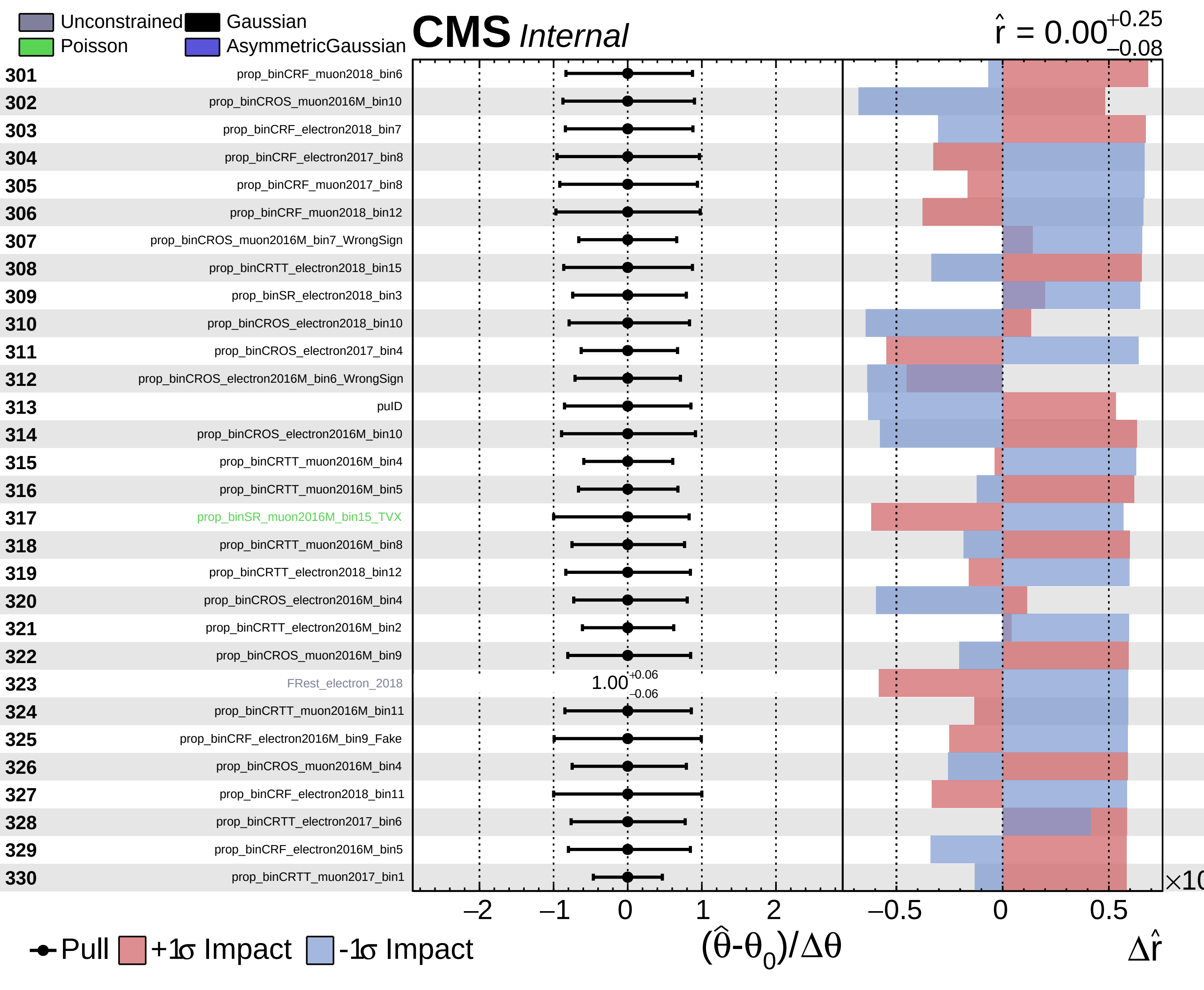

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
211 prop_binSR_muon2017_bin13_TVX
212 prop_binSR_muon2016M_bin3
213 prop_binCRF_muon2018_bin11
214 prop_binCRTT_electron2017_bin4
215 prop_binSR_muon2018_bin3
216 prop_binCRTT_muon2016M_bin2
217 prop_binCRF_muon2018_bin7
218 prop_binCROS_electron2016M_bin9_WrongSign
219 prop_binCRTT_muon2017_bin5
220 prop_binSR_muon2018_bin4
221 prop_binCRTT_electron2016M_bin6
222 prop_binCRF_electron2017_bin3
223 prop_binCRF_electron2018_bin8
224 prop_binCRTT_electron2017_bin3
225 TES
226 pu
227 prop_binCRTT_electron2018_bin3
228 prop_binCROS_muon2016M_bin5
229 prop_binCRF_electron2018_bin6
230 prop_binCROS_electron2016M_bin1
231 prop_binCROS_muon2017_bin13
232 jes_2018
233 prop_binSR_electron2017_bin17_TTTo2L2Nu
234 prop_binCROS_muon2017_bin8
235 prop_binSR_muon2017_bin13_Triboson
236 FRest_electron_2017
$1.00^{+0.07}_{-0.07}$
237 prop_binCRF_electron2017_bin12_TVX
238 prop_binCRTT_muon2018_bin12
239 prop_binCRTT_electron2017_bin2
240 prop_binCRTT_electron2018_bin5
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



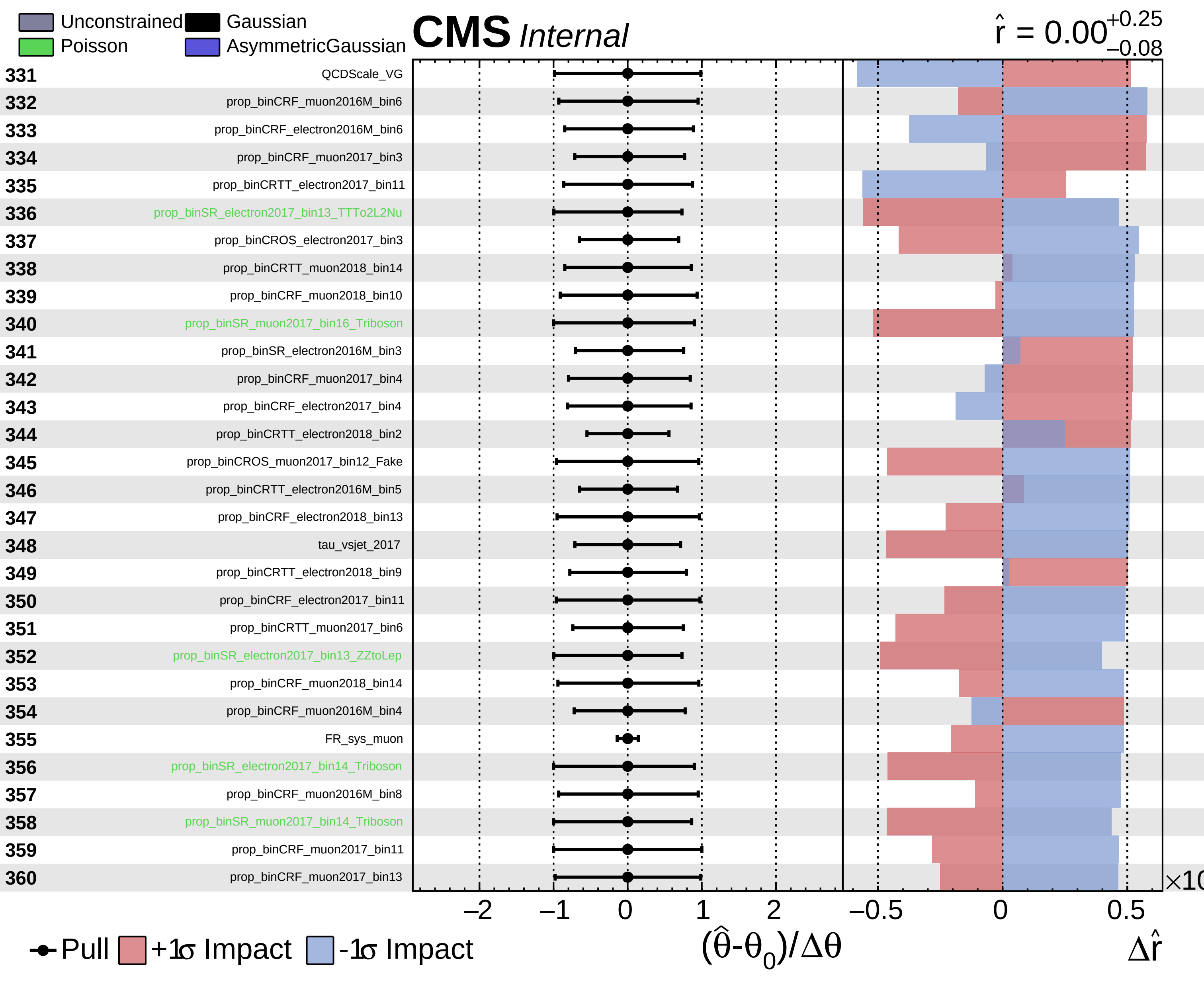
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.25 −0.08
241 prop_binCRF_muon2018_bin9
242 prop_binCRTT_muon2018_bin8
243 prop_binCRTT_electron2017_bin5
244 prop_binCRF_electron2017_bin7
245 prop_binCRTT_electron2017_bin9
246 prop_binCROS_electron2018_bin6_WrongSign
247 prop_binCRTT_muon2018_bin9
248 prop_binSR_electron2017_bin1
249 TTest_muon_2018
1.00 +0.05 −0.05
250 prop_binCROS_muon2017_bin9_Fake
251 prop_binSR_electron2018_bin4
252 prop_binSR_electron2016M_bin15_ZZtoLep
253 prop_binCRF_electron2016M_bin3
254 prop_binCRF_electron2018_bin10
255 prop_binCRF_electron2016M_bin9_TTTo2L2Nu
256 prop_binCRTT_muon2018_bin2
257 prop_binCRTT_electron2016M_bin1
258 prop_binCRF_muon2017_bin10
259 prop_binCRTT_muon2017_bin4
260 prop_binSR_muon2016M_bin2
261 prop_binSR_electron2017_bin4
262 prop_binSR_muon2018_bin15_Triboson
263 prop_binCROS_electron2017_bin9_WrongSign
264 prop_binCROS_electron2018_bin5
265 prop_binCRTT_muon2017_bin17
266 prop_binCRF_muon2016M_bin7
267 prop_binCRTT_muon2017_bin7
268 prop_binCRTT_muon2018_bin13
269 prop_binCRTT_electron2017_bin1
270 prop_binCRTT_electron2018_bin8
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ₀)/Δθ
Δr̂



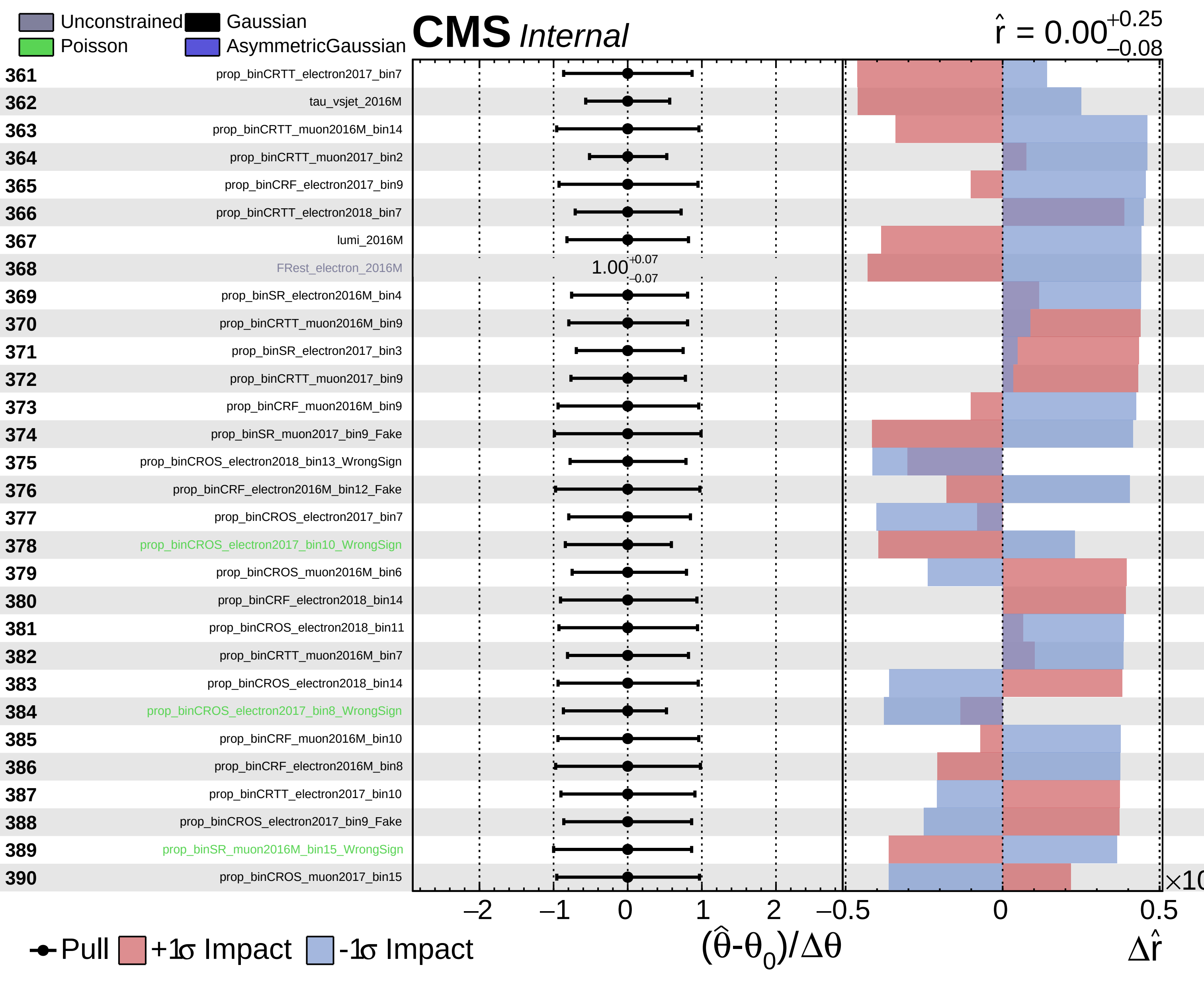
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
271 prop_binCROS_muon2018_bin10
272 prop_binCROS_muon2018_bin13_WrongSign
273 prop_binCRF_electron2017_bin12_Fake
274 prop_binCRTT_muon2017_bin10
275 prop_binCRTT_muon2018_bin10
276 prop_binCRTT_muon2018_bin7
277 prop_binCROS_electron2017_bin6
278 prop_binCRTT_muon2018_bin5
279 prop_binSR_muon2018_bin15_TVX
280 prop_binCRTT_muon2018_bin11
281 prop_binCROS_electron2016M_bin5
282 prop_binCRF_muon2017_bin9
283 prop_binCRTT_muon2016M_bin10
284 prop_binCRF_electron2018_bin12
285 prop_binCRF_electron2018_bin15_TTTo2L2Nu
286 prop_binCROS_muon2018_bin12
287 lumi_2017
288 prop_binCRTT_electron2017_bin8
289 prop_binCROS_muon2018_bin8
290 prop_binCRF_muon2017_bin12_Fake
291 prop_binCRF_electron2018_bin9
292 prop_binSR_muon2016M_bin16_ZZtoLep
293 prop_binCRTT_muon2017_bin8
294 prop_binCRF_muon2016M_bin11_ZZtoLep
295 FRest_muon_2018
$1.00^{+0.06}_{-0.06}$
296 prop_binCRTT_electron2018_bin10
297 FRest_muon_2016M
$1.00^{+0.07}_{-0.07}$
298 prop_binSR_electron2018_bin17_Triboson
299 prop_binCRF_muon2018_bin13
300 prop_binCRTT_muon2017_bin15
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

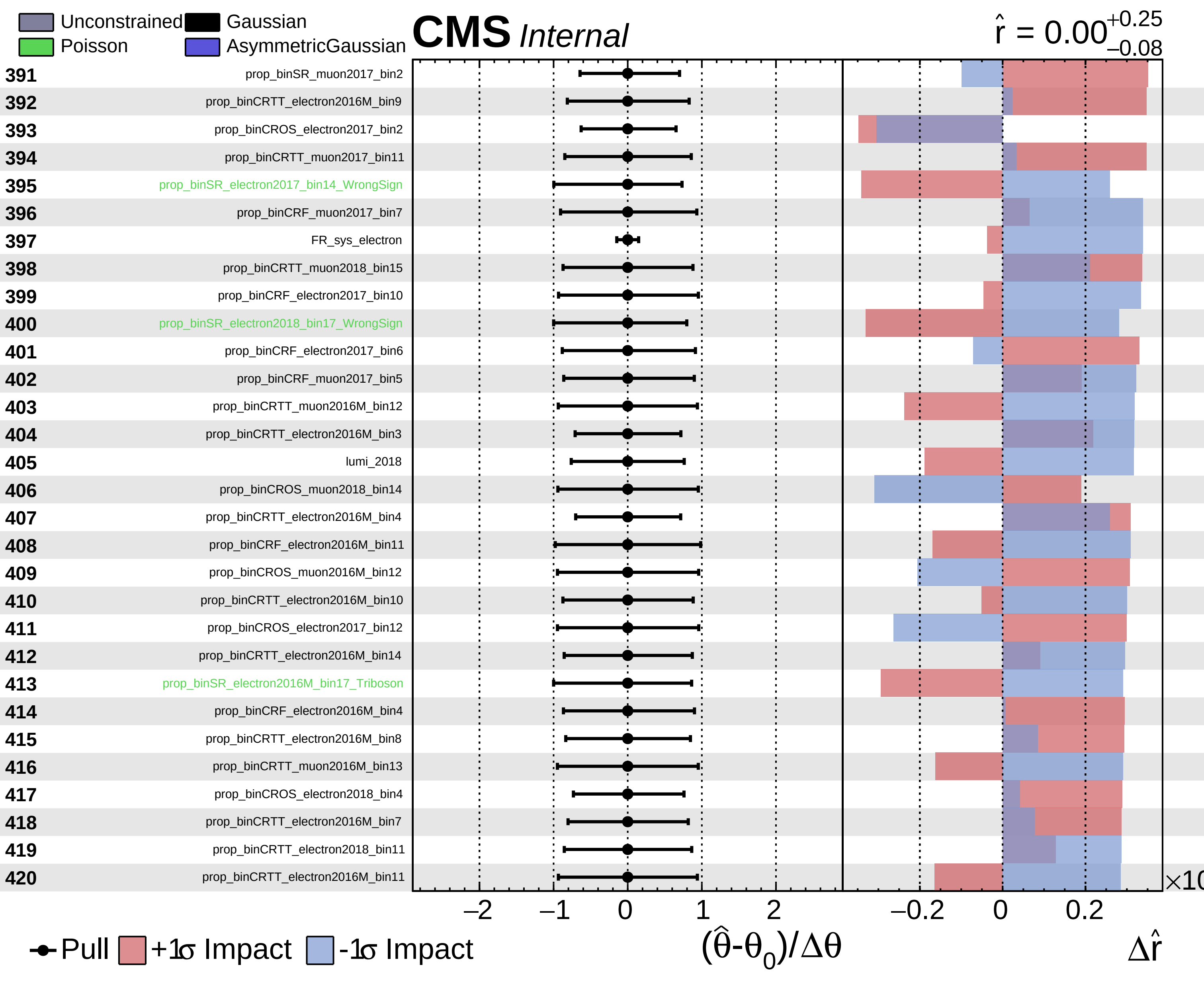


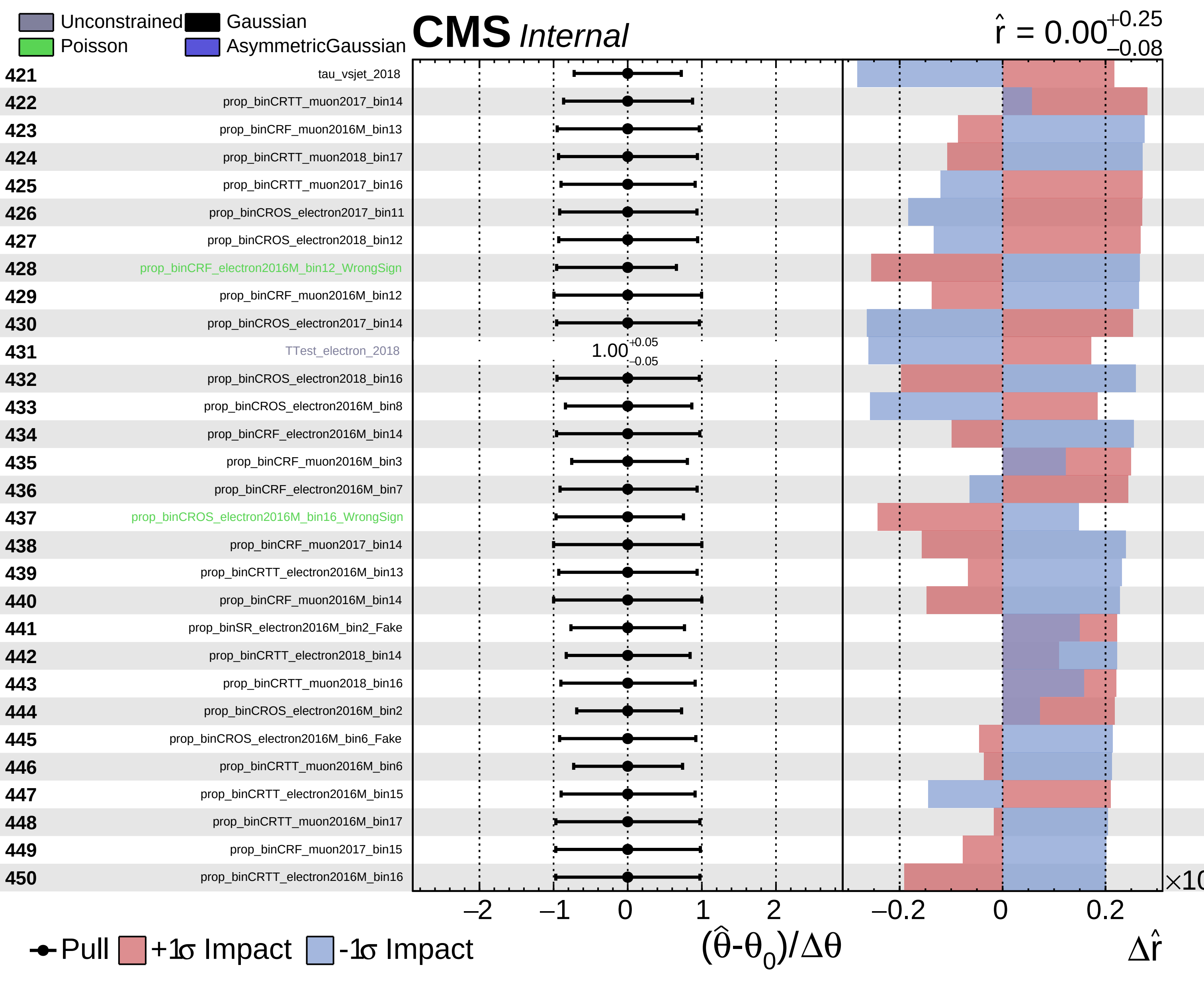
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
301 prop_binCRF_muon2018_bin6
302 prop_binCROS_muon2016M_bin10
303 prop_binCRF_electron2018_bin7
304 prop_binCRF_electron2017_bin8
305 prop_binCRF_muon2017_bin8
306 prop_binCRF_muon2018_bin12
307 prop_binCROS_muon2016M_bin7_WrongSign
308 prop_binCRTT_electron2018_bin15
309 prop_binSR_electron2018_bin3
310 prop_binCROS_electron2018_bin10
311 prop_binCROS_electron2017_bin4
312 prop_binCROS_electron2016M_bin6_WrongSign
313 puID
314 prop_binCROS_electron2016M_bin10
315 prop_binCRTT_muon2016M_bin4
316 prop_binCRTT_muon2016M_bin5
317 prop_binSR_muon2016M_bin15_TVX
318 prop_binCRTT_muon2016M_bin8
319 prop_binCRTT_electron2018_bin12
320 prop_binCROS_electron2016M_bin4
321 prop_binCRTT_electron2016M_bin2
322 prop_binCROS_muon2016M_bin9
323 FRest_electron_2018
$1.00^{+0.06}_{-0.06}$
324 prop_binCRTT_muon2016M_bin11
325 prop_binCRF_electron2016M_bin9_Fake
326 prop_binCROS_muon2016M_bin4
327 prop_binCRF_electron2018_bin11
328 prop_binCRTT_electron2017_bin6
329 prop_binCRF_electron2016M_bin5
330 prop_binCRTT_muon2017_bin1
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



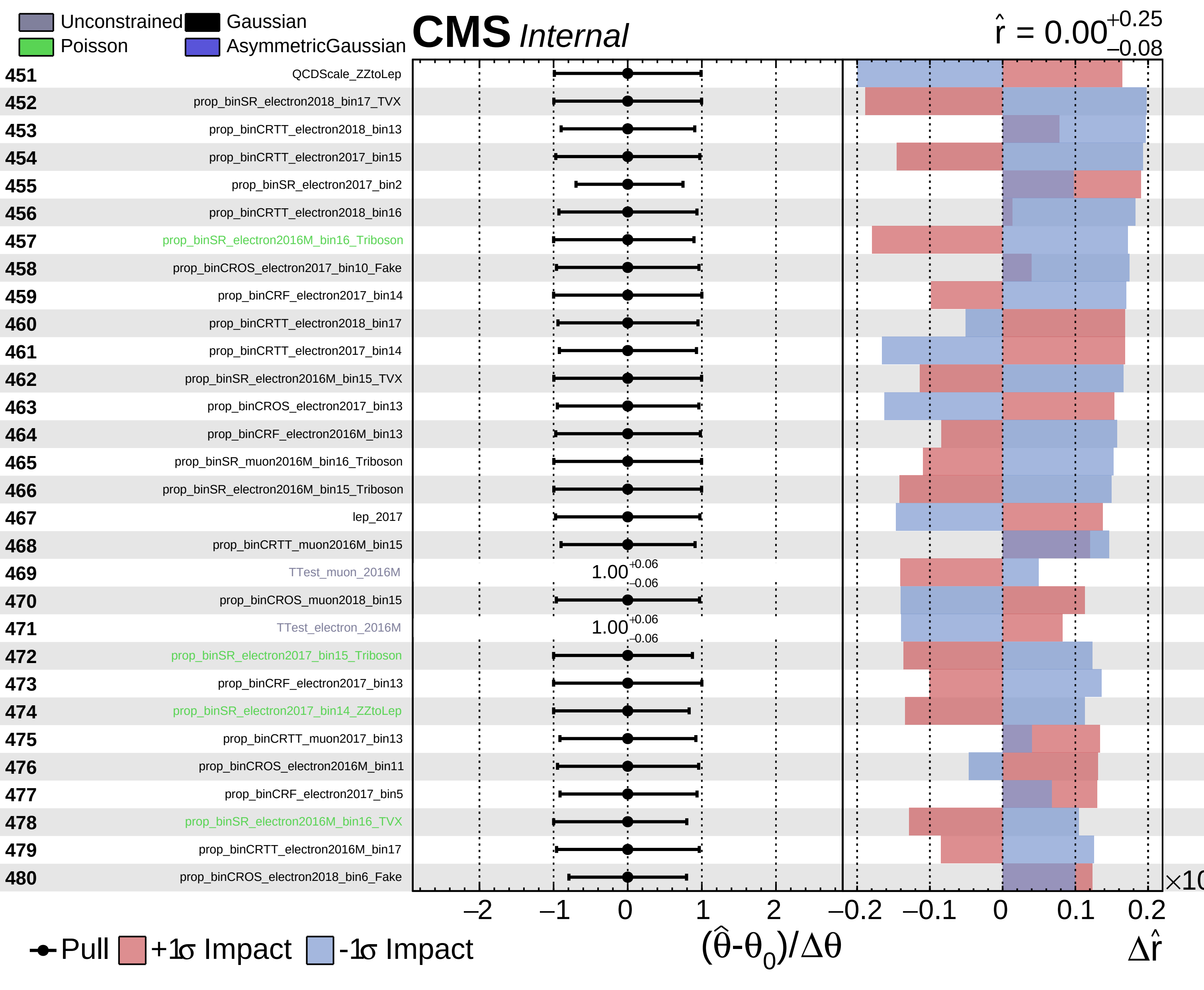
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
331 QCDScale_VG
332 prop_binCRF_muon2016M_bin6
333 prop_binCRF_electron2016M_bin6
334 prop_binCRF_muon2017_bin3
335 prop_binCRTT_electron2017_bin11
336 prop_binSR_electron2017_bin13_TTTo2L2Nu
337 prop_binCROS_electron2017_bin3
338 prop_binCRTT_muon2018_bin14
339 prop_binCRF_muon2018_bin10
340 prop_binSR_muon2017_bin16_Triboson
341 prop_binSR_electron2016M_bin3
342 prop_binCRF_muon2017_bin4
343 prop_binCRF_electron2017_bin4
344 prop_binCRTT_electron2018_bin2
345 prop_binCROS_muon2017_bin12_Fake
346 prop_binCRTT_electron2016M_bin5
347 prop_binCRF_electron2018_bin13
348 tau_vsjet_2017
349 prop_binCRTT_electron2018_bin9
350 prop_binCRF_electron2017_bin11
351 prop_binCRTT_muon2017_bin6
352 prop_binSR_electron2017_bin13_ZZtoLep
353 prop_binCRF_muon2018_bin14
354 prop_binCRF_muon2016M_bin4
355 FR_sys_muon
356 prop_binSR_electron2017_bin14_Triboson
357 prop_binCRF_muon2016M_bin8
358 prop_binSR_muon2017_bin14_Triboson
359 prop_binCRF_muon2017_bin11
360 prop_binCRF_muon2017_bin13
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.25 −0.08
361 prop_binCRTT_electron2017_bin7
362 tau_vsjet_2016M
363 prop_binCRTT_muon2016M_bin14
364 prop_binCRTT_muon2017_bin2
365 prop_binCRF_electron2017_bin9
366 prop_binCRTT_electron2018_bin7
367 lumi_2016M
368 FRest_electron_2016M
1.00 +0.07 −0.07
369 prop_binSR_electron2016M_bin4
370 prop_binCRTT_muon2016M_bin9
371 prop_binSR_electron2017_bin3
372 prop_binCRTT_muon2017_bin9
373 prop_binCRF_muon2016M_bin9
374 prop_binSR_muon2017_bin9_Fake
375 prop_binCROS_electron2018_bin13_WrongSign
376 prop_binCRF_electron2016M_bin12_Fake
377 prop_binCROS_electron2017_bin7
378 prop_binCROS_electron2017_bin10_WrongSign
379 prop_binCROS_muon2016M_bin6
380 prop_binCRF_electron2018_bin14
381 prop_binCROS_electron2018_bin11
382 prop_binCRTT_muon2016M_bin7
383 prop_binCROS_electron2018_bin14
384 prop_binCROS_electron2017_bin8_WrongSign
385 prop_binCRF_muon2016M_bin10
386 prop_binCRF_electron2016M_bin8
387 prop_binCRTT_electron2017_bin10
388 prop_binCROS_electron2017_bin9_Fake
389 prop_binSR_muon2016M_bin15_WrongSign
390 prop_binCROS_muon2017_bin15
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

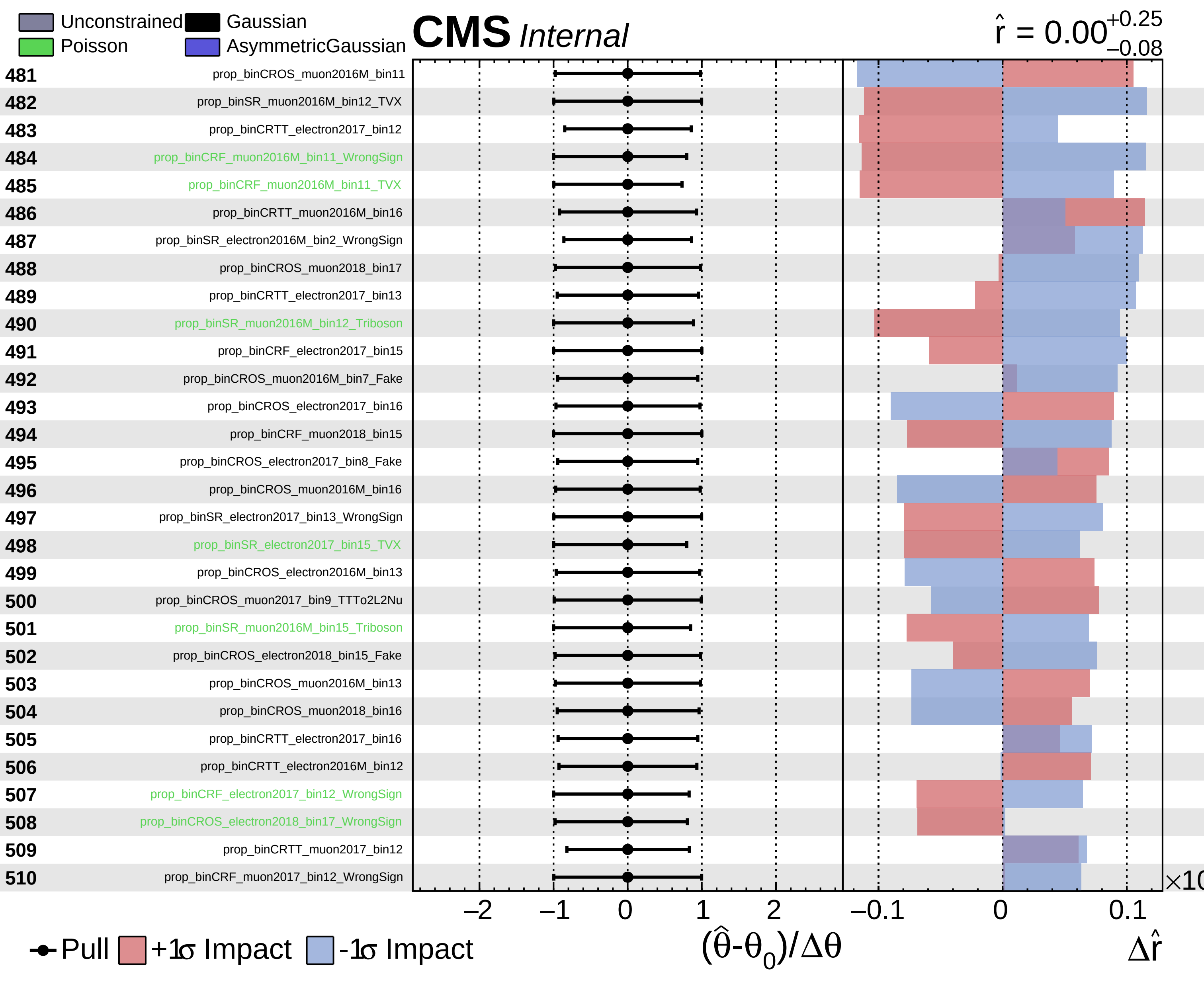

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
391 prop_binSR_muon2017_bin2
392 prop_binCRTT_electron2016M_bin9
393 prop_binCROS_electron2017_bin2
394 prop_binCRTT_muon2017_bin11
395 prop_binSR_electron2017_bin14_WrongSign
396 prop_binCRF_muon2017_bin7
397 FR_sys_electron
398 prop_binCRTT_muon2018_bin15
399 prop_binCRF_electron2017_bin10
400 prop_binSR_electron2018_bin17_WrongSign
401 prop_binCRF_electron2017_bin6
402 prop_binCRF_muon2017_bin5
403 prop_binCRTT_muon2016M_bin12
404 prop_binCRTT_electron2016M_bin3
405 lumi_2018
406 prop_binCROS_muon2018_bin14
407 prop_binCRTT_electron2016M_bin4
408 prop_binCRF_electron2016M_bin11
409 prop_binCROS_muon2016M_bin12
410 prop_binCRTT_electron2016M_bin10
411 prop_binCROS_electron2017_bin12
412 prop_binCRTT_electron2016M_bin14
413 prop_binSR_electron2016M_bin17_Triboson
414 prop_binCRF_electron2016M_bin4
415 prop_binCRTT_electron2016M_bin8
416 prop_binCRTT_muon2016M_bin13
417 prop_binCROS_electron2018_bin4
418 prop_binCRTT_electron2016M_bin7
419 prop_binCRTT_electron2018_bin11
420 prop_binCRTT_electron2016M_bin11
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



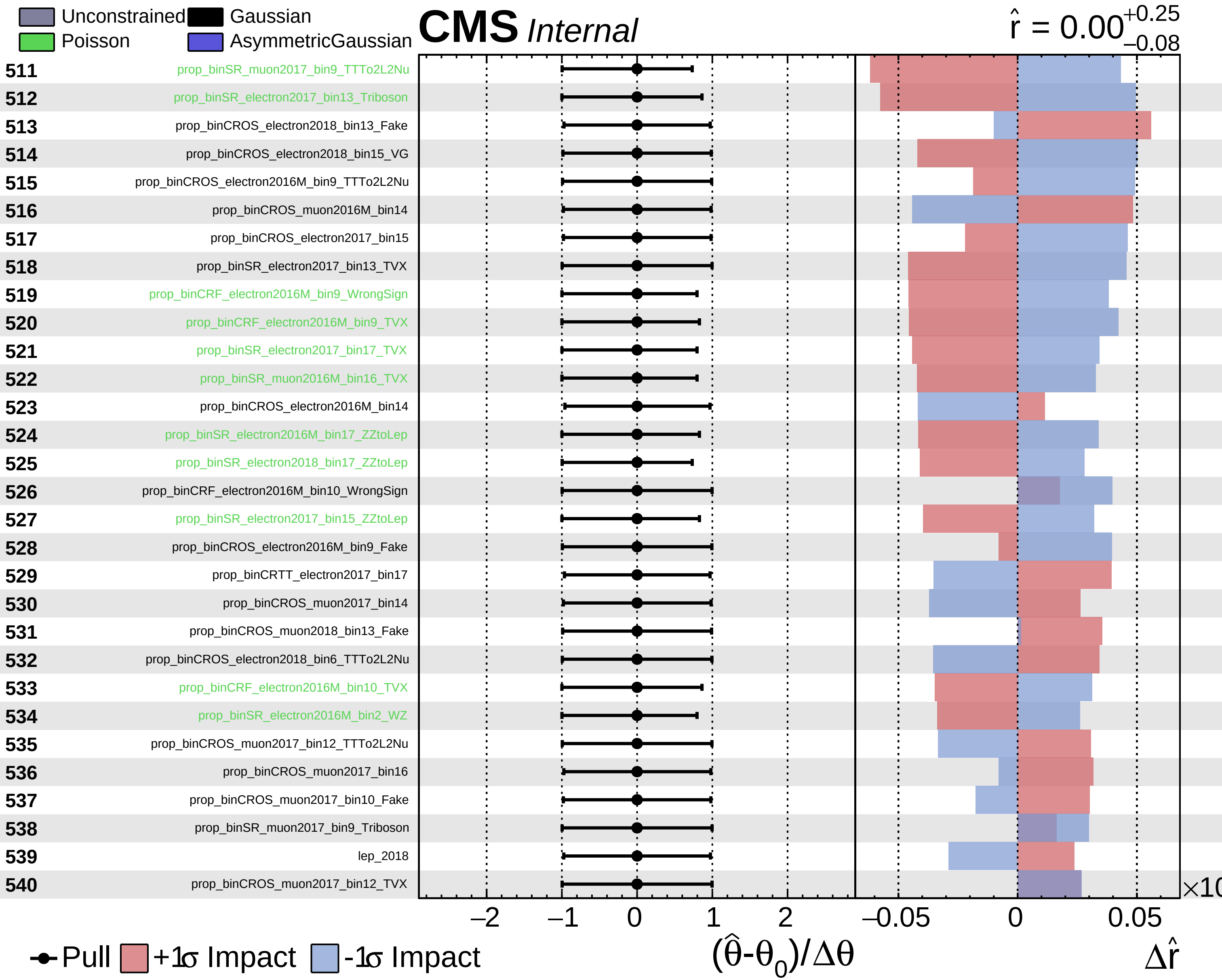
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.25 −0.08
421 tau_vsjet_2018
422 prop_binCRTT_muon2017_bin14
423 prop_binCRF_muon2016M_bin13
424 prop_binCRTT_muon2018_bin17
425 prop_binCRTT_muon2017_bin16
426 prop_binCROS_electron2017_bin11
427 prop_binCROS_electron2018_bin12
428 prop_binCRF_electron2016M_bin12_WrongSign
429 prop_binCRF_muon2016M_bin12
430 prop_binCROS_electron2017_bin14
431 TTest_electron_2018
1.00 +0.05 −0.05
432 prop_binCROS_electron2018_bin16
433 prop_binCROS_electron2016M_bin8
434 prop_binCRF_electron2016M_bin14
435 prop_binCRF_muon2016M_bin3
436 prop_binCRF_electron2016M_bin7
437 prop_binCROS_electron2016M_bin16_WrongSign
438 prop_binCRF_muon2017_bin14
439 prop_binCRTT_electron2016M_bin13
440 prop_binCRF_muon2016M_bin14
441 prop_binSR_electron2016M_bin2_Fake
442 prop_binCRTT_electron2018_bin14
443 prop_binCRTT_muon2018_bin16
444 prop_binCROS_electron2016M_bin2
445 prop_binCROS_electron2016M_bin6_Fake
446 prop_binCRTT_muon2016M_bin6
447 prop_binCRTT_electron2016M_bin15
448 prop_binCRTT_muon2016M_bin17
449 prop_binCRF_muon2017_bin15
450 prop_binCRTT_electron2016M_bin16
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



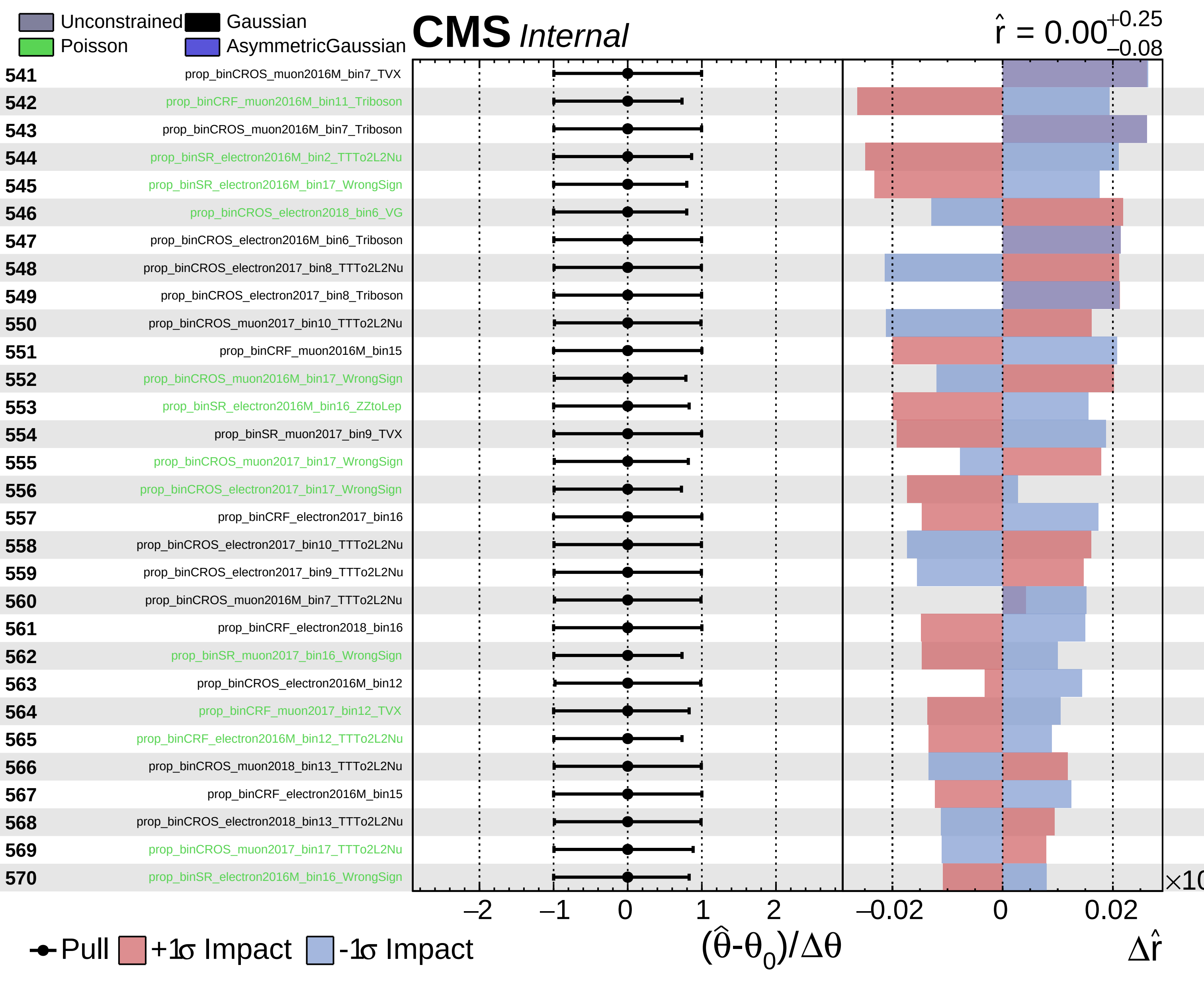
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
451 QCDScale_ZZtoLep
452 prop_binSR_electron2018_bin17_TVX
453 prop_binCRTT_electron2018_bin13
454 prop_binCRTT_electron2017_bin15
455 prop_binSR_electron2017_bin2
456 prop_binCRTT_electron2018_bin16
457 prop_binSR_electron2016M_bin16_Triboson
458 prop_binCROS_electron2017_bin10_Fake
459 prop_binCRF_electron2017_bin14
460 prop_binCRTT_electron2018_bin17
461 prop_binCRTT_electron2017_bin14
462 prop_binSR_electron2016M_bin15_TVX
463 prop_binCROS_electron2017_bin13
464 prop_binCRF_electron2016M_bin13
465 prop_binSR_muon2016M_bin16_Triboson
466 prop_binSR_electron2016M_bin15_Triboson
467 lep_2017
468 prop_binCRTT_muon2016M_bin15
469 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
470 prop_binCROS_muon2018_bin15
471 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
472 prop_binSR_electron2017_bin15_Triboson
473 prop_binCRF_electron2017_bin13
474 prop_binSR_electron2017_bin14_ZZtoLep
475 prop_binCRTT_muon2017_bin13
476 prop_binCROS_electron2016M_bin11
477 prop_binCRF_electron2017_bin5
478 prop_binSR_electron2016M_bin16_TVX
479 prop_binCRTT_electron2016M_bin17
480 prop_binCROS_electron2018_bin6_Fake
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



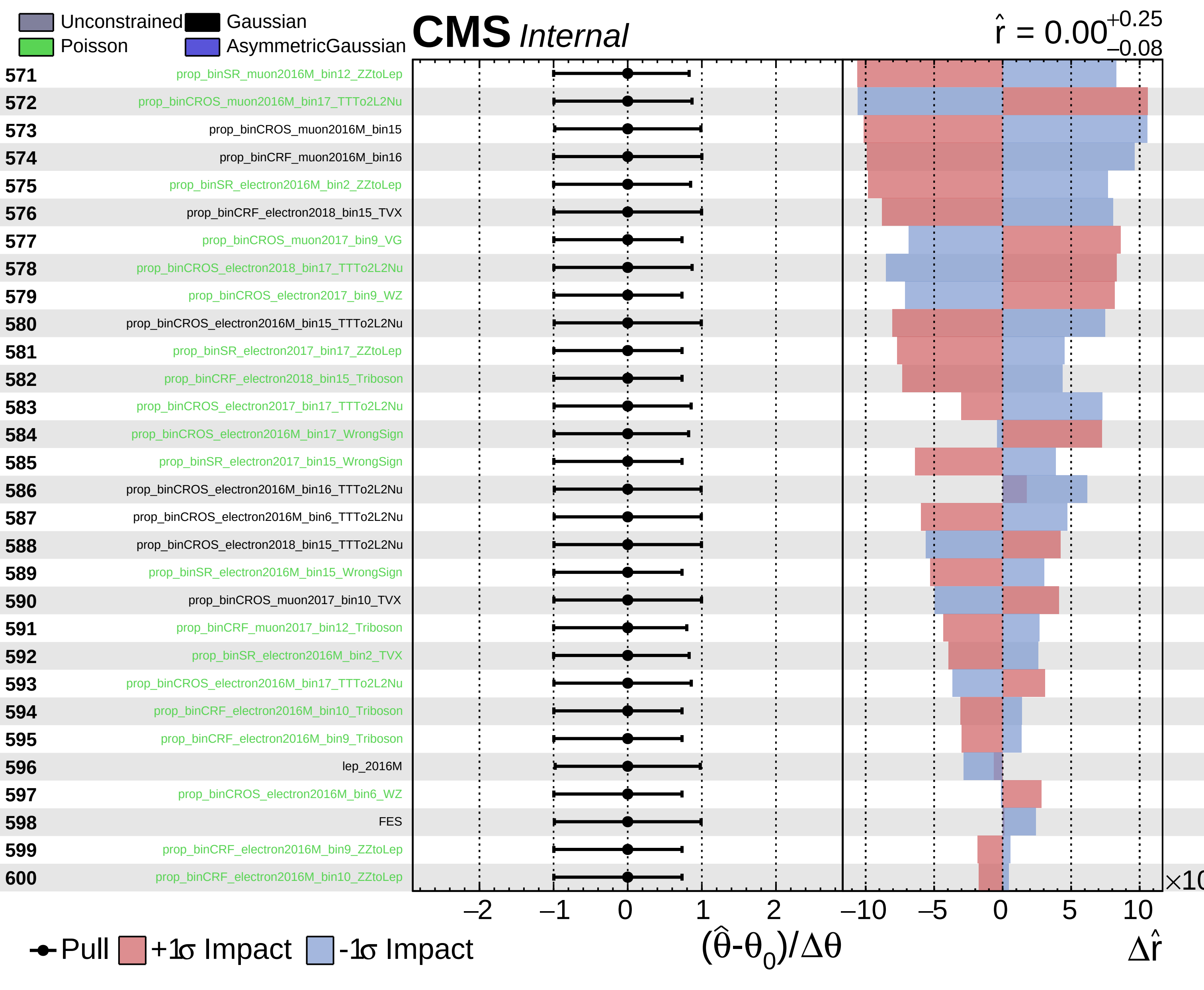
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
481 prop_binCROS_muon2016M_bin11
482 prop_binSR_muon2016M_bin12_TVX
483 prop_binCRTT_electron2017_bin12
484 prop_binCRF_muon2016M_bin11_WrongSign
485 prop_binCRF_muon2016M_bin11_TVX
486 prop_binCRTT_muon2016M_bin16
487 prop_binSR_electron2016M_bin2_WrongSign
488 prop_binCROS_muon2018_bin17
489 prop_binCRTT_electron2017_bin13
490 prop_binSR_muon2016M_bin12_Triboson
491 prop_binCRF_electron2017_bin15
492 prop_binCROS_muon2016M_bin7_Fake
493 prop_binCROS_electron2017_bin16
494 prop_binCRF_muon2018_bin15
495 prop_binCROS_electron2017_bin8_Fake
496 prop_binCROS_muon2016M_bin16
497 prop_binSR_electron2017_bin13_WrongSign
498 prop_binSR_electron2017_bin15_TVX
499 prop_binCROS_electron2016M_bin13
500 prop_binCROS_muon2017_bin9_TTTo2L2Nu
501 prop_binSR_muon2016M_bin15_Triboson
502 prop_binCROS_electron2018_bin15_Fake
503 prop_binCROS_muon2016M_bin13
504 prop_binCROS_muon2018_bin16
505 prop_binCRTT_electron2017_bin16
506 prop_binCRTT_electron2016M_bin12
507 prop_binCRF_electron2017_bin12_WrongSign
508 prop_binCROS_electron2018_bin17_WrongSign
509 prop_binCRTT_muon2017_bin12
510 prop_binCRF_muon2017_bin12_WrongSign
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



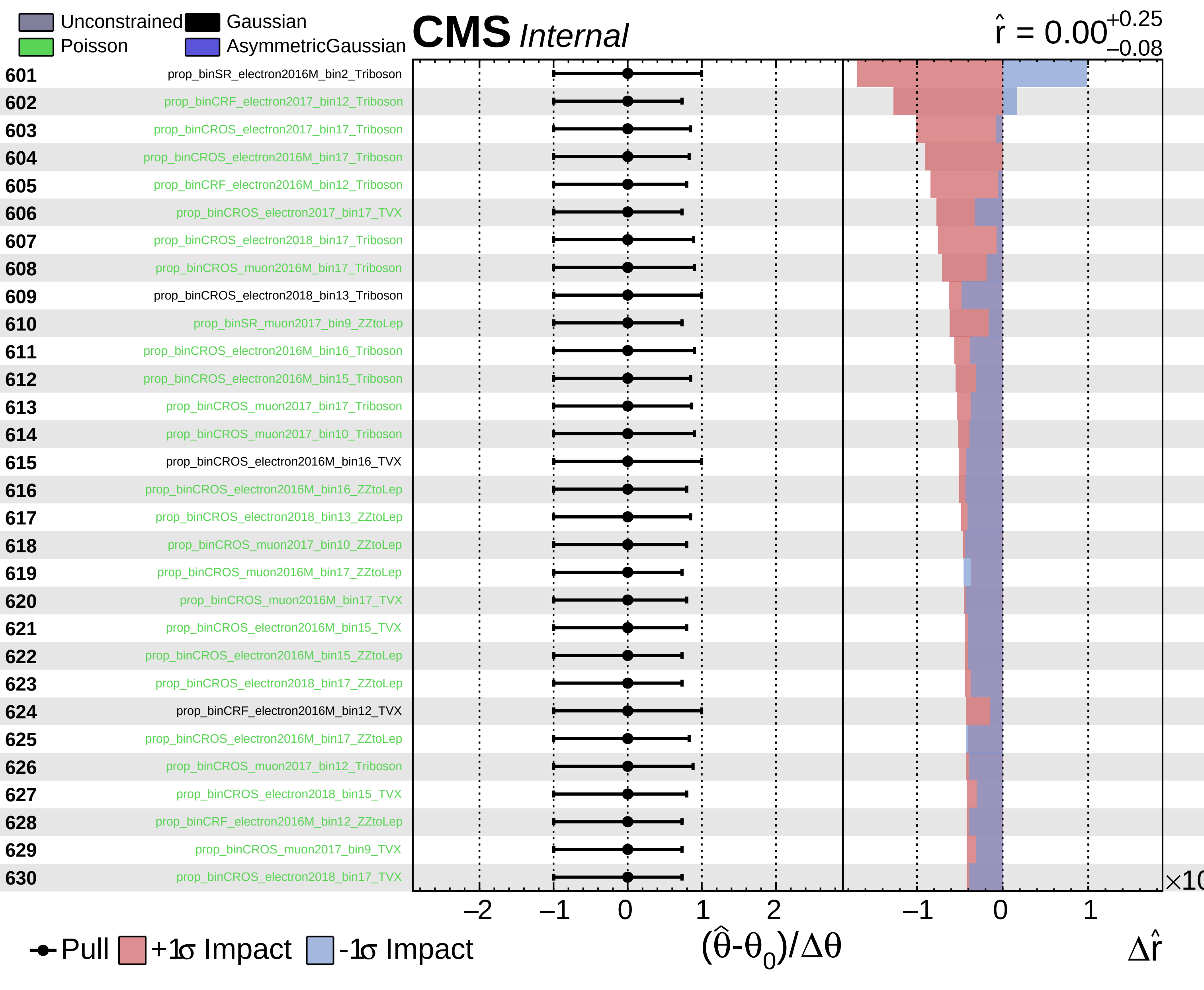
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.25 −0.08
511 prop_binSR_muon2017_bin9_TTTo2L2Nu
512 prop_binSR_electron2017_bin13_Triboson
513 prop_binCROS_electron2018_bin13_Fake
514 prop_binCROS_electron2018_bin15_VG
515 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
516 prop_binCROS_muon2016M_bin14
517 prop_binCROS_electron2017_bin15
518 prop_binSR_electron2017_bin13_TVX
519 prop_binCRF_electron2016M_bin9_WrongSign
520 prop_binCRF_electron2016M_bin9_TVX
521 prop_binSR_electron2017_bin17_TVX
522 prop_binSR_muon2016M_bin16_TVX
523 prop_binCROS_electron2016M_bin14
524 prop_binSR_electron2016M_bin17_ZZtoLep
525 prop_binSR_electron2018_bin17_ZZtoLep
526 prop_binCRF_electron2016M_bin10_WrongSign
527 prop_binSR_electron2017_bin15_ZZtoLep
528 prop_binCROS_electron2016M_bin9_Fake
529 prop_binCRTT_electron2017_bin17
530 prop_binCROS_muon2017_bin14
531 prop_binCROS_muon2018_bin13_Fake
532 prop_binCROS_electron2018_bin6_TTTo2L2Nu
533 prop_binCRF_electron2016M_bin10_TVX
534 prop_binSR_electron2016M_bin2_WZ
535 prop_binCROS_muon2017_bin12_TTTo2L2Nu
536 prop_binCROS_muon2017_bin16
537 prop_binCROS_muon2017_bin10_Fake
538 prop_binSR_muon2017_bin9_Triboson
539 lep_2018
540 prop_binCROS_muon2017_bin12_TVX
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



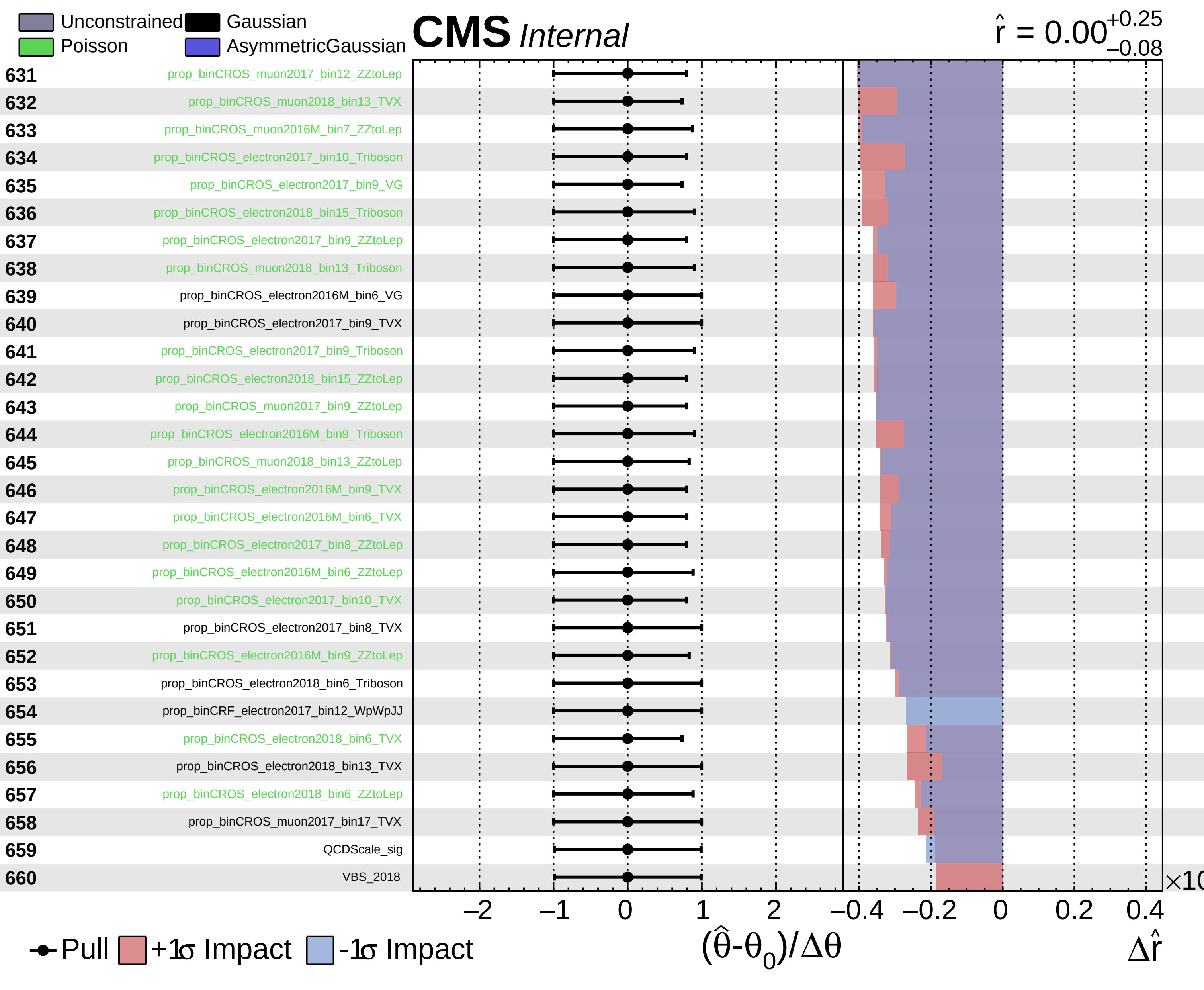
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.25 −0.08
541 prop_binCROS_muon2016M_bin7_TVX
542 prop_binCRF_muon2016M_bin11_Triboson
543 prop_binCROS_muon2016M_bin7_Triboson
544 prop_binSR_electron2016M_bin2_TTTo2L2Nu
545 prop_binSR_electron2016M_bin17_WrongSign
546 prop_binCROS_electron2018_bin6_VG
547 prop_binCROS_electron2016M_bin6_Triboson
548 prop_binCROS_electron2017_bin8_TTTo2L2Nu
549 prop_binCROS_electron2017_bin8_Triboson
550 prop_binCROS_muon2017_bin10_TTTo2L2Nu
551 prop_binCRF_muon2016M_bin15
552 prop_binCROS_muon2016M_bin17_WrongSign
553 prop_binSR_electron2016M_bin16_ZZtoLep
554 prop_binSR_muon2017_bin9_TVX
555 prop_binCROS_muon2017_bin17_WrongSign
556 prop_binCROS_electron2017_bin17_WrongSign
557 prop_binCRF_electron2017_bin16
558 prop_binCROS_electron2017_bin10_TTTo2L2Nu
559 prop_binCROS_electron2017_bin9_TTTo2L2Nu
560 prop_binCROS_muon2016M_bin7_TTTo2L2Nu
561 prop_binCRF_electron2018_bin16
562 prop_binSR_muon2017_bin16_WrongSign
563 prop_binCROS_electron2016M_bin12
564 prop_binCRF_muon2017_bin12_TVX
565 prop_binCRF_electron2016M_bin12_TTTo2L2Nu
566 prop_binCROS_muon2018_bin13_TTTo2L2Nu
567 prop_binCRF_electron2016M_bin15
568 prop_binCROS_electron2018_bin13_TTTo2L2Nu
569 prop_binCROS_muon2017_bin17_TTTo2L2Nu
570 prop_binSR_electron2016M_bin16_WrongSign
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

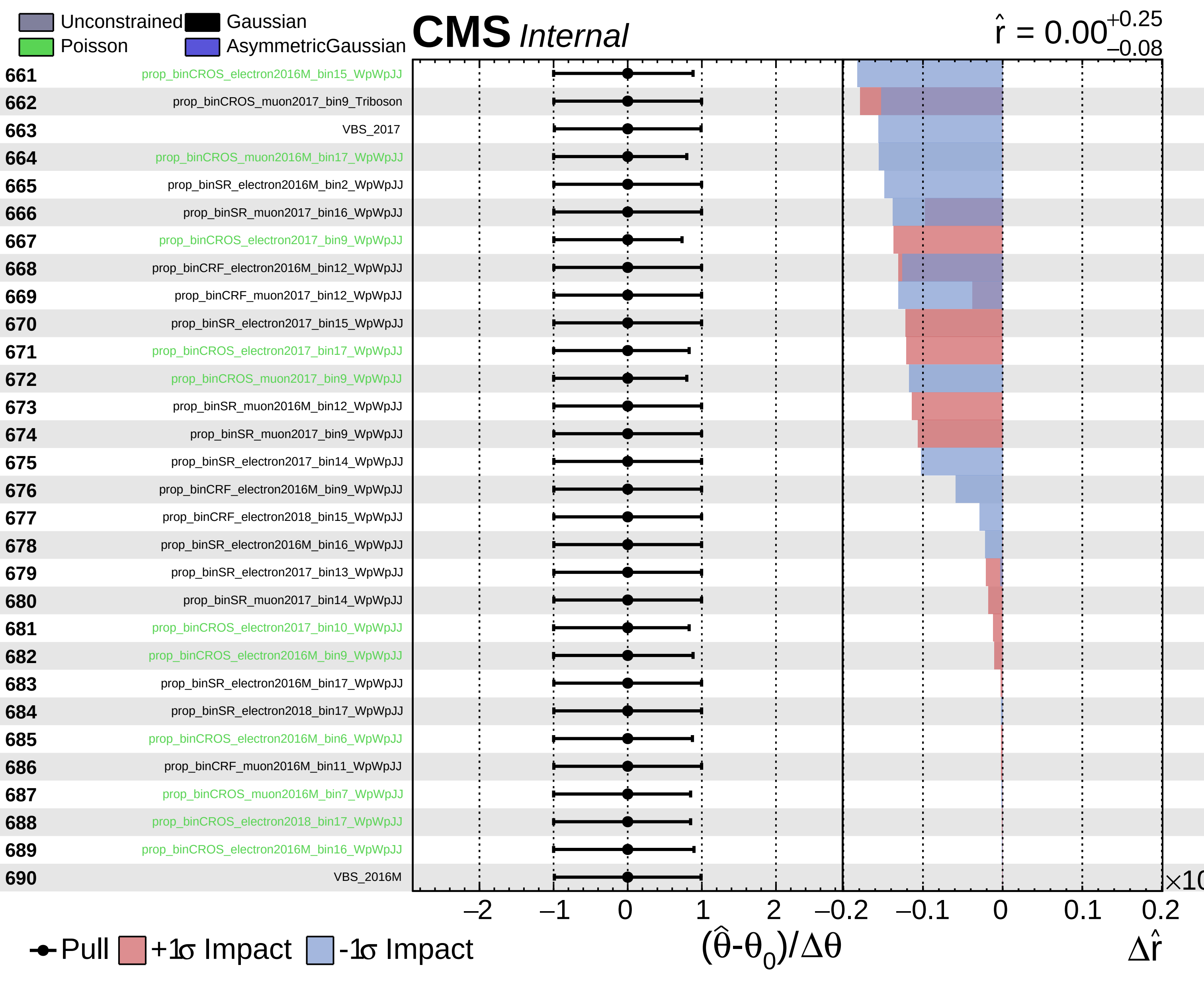


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
571 prop_binSR_muon2016M_bin12_ZZtoLep
572 prop_binCROS_muon2016M_bin17_TTTo2L2Nu
573 prop_binCROS_muon2016M_bin15
574 prop_binCRF_muon2016M_bin16
575 prop_binSR_electron2016M_bin2_ZZtoLep
576 prop_binCRF_electron2018_bin15_TVX
577 prop_binCROS_muon2017_bin9_VG
578 prop_binCROS_electron2018_bin17_TTTo2L2Nu
579 prop_binCROS_electron2017_bin9_WZ
580 prop_binCROS_electron2016M_bin15_TTTo2L2Nu
581 prop_binSR_electron2017_bin17_ZZtoLep
582 prop_binCRF_electron2018_bin15_Triboson
583 prop_binCROS_electron2017_bin17_TTTo2L2Nu
584 prop_binCROS_electron2016M_bin17_WrongSign
585 prop_binSR_electron2017_bin15_WrongSign
586 prop_binCROS_electron2016M_bin16_TTTo2L2Nu
587 prop_binCROS_electron2016M_bin6_TTTo2L2Nu
588 prop_binCROS_electron2018_bin15_TTTo2L2Nu
589 prop_binSR_electron2016M_bin15_WrongSign
590 prop_binCROS_muon2017_bin10_TVX
591 prop_binCRF_muon2017_bin12_Triboson
592 prop_binSR_electron2016M_bin2_TVX
593 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
594 prop_binCRF_electron2016M_bin10_Triboson
595 prop_binCRF_electron2016M_bin9_Triboson
596 lep_2016M
597 prop_binCROS_electron2016M_bin6_WZ
598 FES
599 prop_binCRF_electron2016M_bin9_ZZtoLep
600 prop_binCRF_electron2016M_bin10_ZZtoLep
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

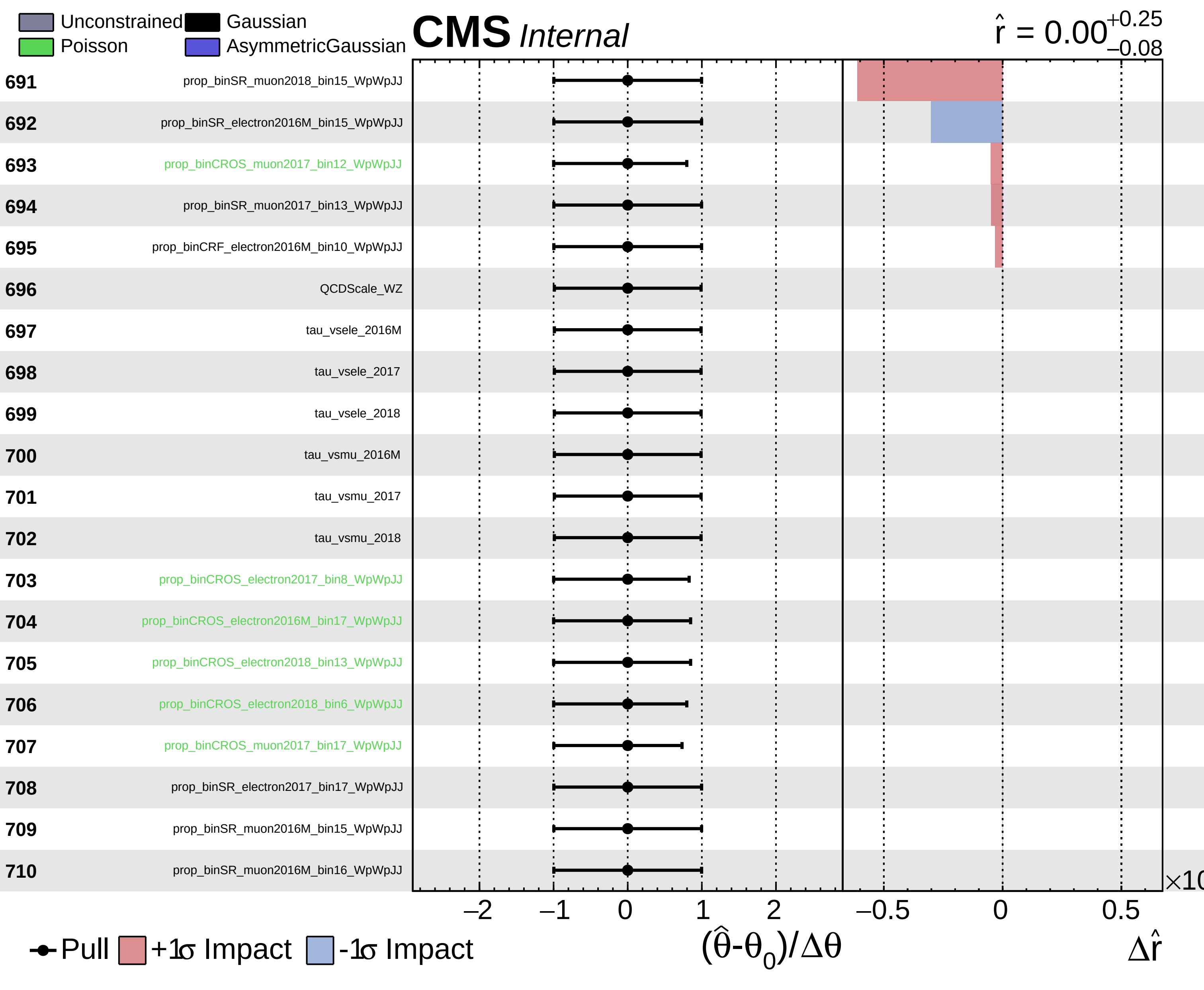


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.25 −0.08
601 prop_binSR_electron2016M_bin2_Triboson
602 prop_binCRF_electron2017_bin12_Triboson
603 prop_binCROS_electron2017_bin17_Triboson
604 prop_binCROS_electron2016M_bin17_Triboson
605 prop_binCRF_electron2016M_bin12_Triboson
606 prop_binCROS_electron2017_bin17_TVX
607 prop_binCROS_electron2018_bin17_Triboson
608 prop_binCROS_muon2016M_bin17_Triboson
609 prop_binCROS_electron2018_bin13_Triboson
610 prop_binSR_muon2017_bin9_ZZtoLep
611 prop_binCROS_electron2016M_bin16_Triboson
612 prop_binCROS_electron2016M_bin15_Triboson
613 prop_binCROS_muon2017_bin17_Triboson
614 prop_binCROS_muon2017_bin10_Triboson
615 prop_binCROS_electron2016M_bin16_TVX
616 prop_binCROS_electron2016M_bin16_ZZtoLep
617 prop_binCROS_electron2018_bin13_ZZtoLep
618 prop_binCROS_muon2017_bin10_ZZtoLep
619 prop_binCROS_muon2016M_bin17_ZZtoLep
620 prop_binCROS_muon2016M_bin17_TVX
621 prop_binCROS_electron2016M_bin15_TVX
622 prop_binCROS_electron2016M_bin15_ZZtoLep
623 prop_binCROS_electron2018_bin17_ZZtoLep
624 prop_binCRF_electron2016M_bin12_TVX
625 prop_binCROS_electron2016M_bin17_ZZtoLep
626 prop_binCROS_muon2017_bin12_Triboson
627 prop_binCROS_electron2018_bin15_TVX
628 prop_binCRF_electron2016M_bin12_ZZtoLep
629 prop_binCROS_muon2017_bin9_TVX
630 prop_binCROS_electron2018_bin17_TVX
−2 −1 0 1 2
−1 0 1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
631 prop_binCROS_muon2017_bin12_ZZtoLep
632 prop_binCROS_muon2018_bin13_TVX
633 prop_binCROS_muon2016M_bin7_ZZtoLep
634 prop_binCROS_electron2017_bin10_Triboson
635 prop_binCROS_electron2017_bin9_VG
636 prop_binCROS_electron2018_bin15_Triboson
637 prop_binCROS_electron2017_bin9_ZZtoLep
638 prop_binCROS_muon2018_bin13_Triboson
639 prop_binCROS_electron2016M_bin6_VG
640 prop_binCROS_electron2017_bin9_TVX
641 prop_binCROS_electron2017_bin9_Triboson
642 prop_binCROS_electron2018_bin15_ZZtoLep
643 prop_binCROS_muon2017_bin9_ZZtoLep
644 prop_binCROS_electron2016M_bin9_Triboson
645 prop_binCROS_muon2018_bin13_ZZtoLep
646 prop_binCROS_electron2016M_bin9_TVX
647 prop_binCROS_electron2016M_bin6_TVX
648 prop_binCROS_electron2017_bin8_ZZtoLep
649 prop_binCROS_electron2016M_bin6_ZZtoLep
650 prop_binCROS_electron2017_bin10_TVX
651 prop_binCROS_electron2017_bin8_TVX
652 prop_binCROS_electron2016M_bin9_ZZtoLep
653 prop_binCROS_electron2018_bin6_Triboson
654 prop_binCRF_electron2017_bin12_WpWpJJ
655 prop_binCROS_electron2018_bin6_TVX
656 prop_binCROS_electron2018_bin13_TVX
657 prop_binCROS_electron2018_bin6_ZZtoLep
658 prop_binCROS_muon2017_bin17_TVX
659 QCDScale_sig
660 VBS_2018
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
661 prop_binCROS_electron2016M_bin15_WpWpJJ
662 prop_binCROS_muon2017_bin9_Triboson
663 VBS_2017
664 prop_binCROS_muon2016M_bin17_WpWpJJ
665 prop_binSR_electron2016M_bin2_WpWpJJ
666 prop_binSR_muon2017_bin16_WpWpJJ
667 prop_binCROS_electron2017_bin9_WpWpJJ
668 prop_binCRF_electron2016M_bin12_WpWpJJ
669 prop_binCRF_muon2017_bin12_WpWpJJ
670 prop_binSR_electron2017_bin15_WpWpJJ
671 prop_binCROS_electron2017_bin17_WpWpJJ
672 prop_binCROS_muon2017_bin9_WpWpJJ
673 prop_binSR_muon2016M_bin12_WpWpJJ
674 prop_binSR_muon2017_bin9_WpWpJJ
675 prop_binSR_electron2017_bin14_WpWpJJ
676 prop_binCRF_electron2016M_bin9_WpWpJJ
677 prop_binCRF_electron2018_bin15_WpWpJJ
678 prop_binSR_electron2016M_bin16_WpWpJJ
679 prop_binSR_electron2017_bin13_WpWpJJ
680 prop_binSR_muon2017_bin14_WpWpJJ
681 prop_binCROS_electron2017_bin10_WpWpJJ
682 prop_binCROS_electron2016M_bin9_WpWpJJ
683 prop_binSR_electron2016M_bin17_WpWpJJ
684 prop_binSR_electron2018_bin17_WpWpJJ
685 prop_binCROS_electron2016M_bin6_WpWpJJ
686 prop_binCRF_muon2016M_bin11_WpWpJJ
687 prop_binCROS_muon2016M_bin7_WpWpJJ
688 prop_binCROS_electron2018_bin17_WpWpJJ
689 prop_binCROS_electron2016M_bin16_WpWpJJ
690 VBS_2016M
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.08}$
691 prop_binSR_muon2018_bin15_WpWpJJ
692 prop_binSR_electron2016M_bin15_WpWpJJ
693 prop_binCROS_muon2017_bin12_WpWpJJ
694 prop_binSR_muon2017_bin13_WpWpJJ
695 prop_binCRF_electron2016M_bin10_WpWpJJ
696 QCDScale_WZ
697 tau_vsele_2016M
698 tau_vsele_2017
699 tau_vsele_2018
700 tau_vsmu_2016M
701 tau_vsmu_2017
702 tau_vsmu_2018
703 prop_binCROS_electron2017_bin8_WpWpJJ
704 prop_binCROS_electron2016M_bin17_WpWpJJ
705 prop_binCROS_electron2018_bin13_WpWpJJ
706 prop_binCROS_electron2018_bin6_WpWpJJ
707 prop_binCROS_muon2017_bin17_WpWpJJ
708 prop_binSR_electron2017_bin17_WpWpJJ
709 prop_binSR_muon2016M_bin15_WpWpJJ
710 prop_binSR_muon2016M_bin16_WpWpJJ
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$